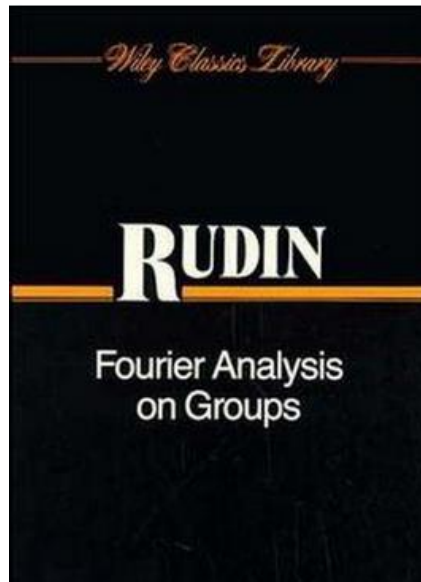


Abstract Harmonic Analysis

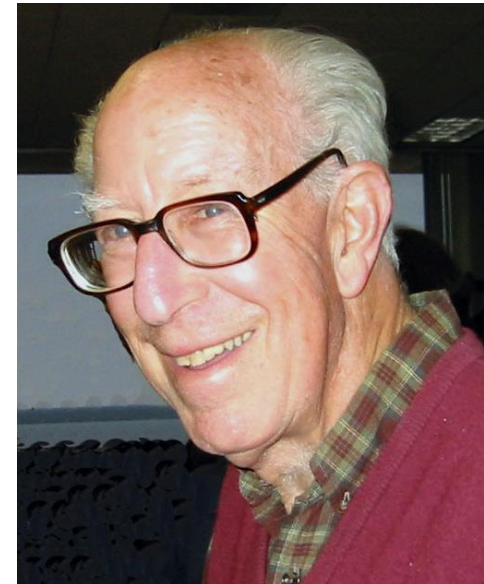


An Appreciation to the Work of

Walter Rudin (1921-2010)

based on his Book

Fourier Analysis on Groups



CHAPTER 1

The Basic Theorems of Fourier Analysis

1. *Haar Measure and Convolution*
2. *The Dual Group and the Fourier Transform*
3. *Fourier Stieltjes Transforms*
4. *Positive Definite Functions*
5. *The Inversion Theorem*
6. *The Plancherel Theorem*
7. *The Pontryagin Duality Theorem*
8. *The Bohr Compactification*
9. *A Characterization of $B(I)$*

The material contained in this chapter forms the core of our subject and is used throughout the later part of this book. Various approaches are possible; the same subject matter is treated, from different points of view, in Cartan and Godement [1], Loomis [1], and Weil [1].

Unless the contrary is explicitly stated, any group mentioned in this book will be abelian and locally compact, with addition as group operation and 0 as identity element (see Appendix B). The abbreviation LCA will be used for „locally compact abelian“.

1. Haar Measure and Convolution (1)

1.1.1. On every LCA group G there exists a non-negative regular measure m (see Appendix E), the so-called *Haar measure* of G , which is not identically 0 and which is *translation-invariant*. That is to say,

$$(1) \quad m(E + x) = m(E)$$

for every $x \in G$ and every Borel set E in G .

For the construction of such a measure, we refer to any of the following standard treatises: Halmos [1], Loomis [1], Montgomery and Zippin [1], and Weil [1]. The idea of the proof is to construct a positive translation-invariant linear functional T on $C_c(G)$, the space of all continuous complex functions on G with compact support. This means that $Tf \geq 0$ if $f \geq 0$ and that $T(f_x) = Tf$, where f_x is the translate of f defined by

$$(2) \quad f_x(y) = f(y - x) \quad (y \in G).$$

As soon as this is done, the Riesz representation theorem shows that there is a measure m with the required properties, such that

$$(3) \quad Tf = \int_G f dm \quad (f \in C_c(G)).$$

1.1.2. If V is a non-empty open subset of G , then $m(V) > 0$. For if $m(V) = 0$ and K is compact, finitely many translates of V cover K , and hence $m(K) = 0$. The regularity of m then implies that $m(E) = 0$ for all Borel sets E in G , a contradiction.

1.1.3. We have spoken of *the* Haar measure of G . This is justified by the following uniqueness theorem:

If m and m' are two Haar measures on G , then $m' = \lambda m$, where λ is a positive constant.

Proof: Fix $g \in C_c(G)$ so that $\int_G g dm = 1$. Define λ by

$$\int_G g(-x) dm'(x) = \lambda.$$

For any $f \in C_c(G)$ we then have

$$\begin{aligned} \int_G f dm' &= \int_G g(y) dm(y) \int_G f(x) dm'(x) \\ &= \int_G g(y) dm(y) \int_G f(x + y) dm'(x) \\ &= \int_G dm'(x) \int_G g(y) f(x + y) dm(y) \\ &= \int_G dm'(x) \int_G g(y - x) f(y) dm(y) \\ &= \int_G f(y) dm(y) \int_G g(y - x) dm'(x) = \lambda \int_G f dm. \end{aligned}$$

1. Haar Measure and Convolution (2)

Hence $m' = \lambda m$. Note that the use of Fubini's theorem was legitimate in the preceding calculation, since the integrands $g(y)f(x+y)$ and $g(y-x)f(y)$ are in $C_c(G \times G)$.

Thus Haar measure is unique, up to a multiplicative positive constant. If G is compact, it is customary to normalize m so that $m(G) = 1$. If G is discrete, any set consisting of a single point is assigned the measure 1. These requirements are of course contradictory if G is a finite group, but this will cause us no difficulty.

Having established the uniqueness of m , we shall now change our notation, and write $\int_G f(x)dx$ in place of $\int_G f dm$. Thus $dx, dy, \dots$ will always denote integration with respect to Haar measure.

1.1.4. For any Borel set E in G , $m(-E) = m(E)$. For if we set $m'(E) = m(-E)$, m' is a Haar measure on G , and so there is a constant λ such that $m(-E) = \lambda m(E)$ for all Borel sets E . Taking E so that $-E = E$, we see that $\lambda = 1$.

1.1.5. Translation in $L^p(G)$. If G is a LCA group and $1 \leq p \leq \infty$, we shall write $L^p(G)$ in place of $L^p(m)$ (see Appendix E7). It is clear that the L^p -norms are translation invariant, i.e., that

$$(1) \quad \|f_x\|_p = \|f\|_p \quad (x \in G),$$

where, we recall, f_x is the translate of f defined by

$$(2) \quad f_x(y) = f(y-x) \quad (y \in G).$$

THEOREM. Suppose $1 \leq p < \infty$ and $f \in L^p(G)$. The map

$$(3) \quad x \rightarrow f_x$$

is a uniformly continuous map of G into $L^p(G)$.

Proof: Let $\varepsilon > 0$ be given. Since $C_c(G)$ is dense in $L^p(G)$ (Appendix E8) there exists $g \in C_c(G)$, with compact support K , such that $\|g - f\|_p < \varepsilon/3$, and the uniform continuity of g (Appendix B9) implies that there is a neighborhood V of 0 in G such that

$$(4) \quad \|g - g_x\|_\infty < \frac{\varepsilon}{3} [m(K)]^{-1/p}$$

for all $x \in V$. Hence $\|g - g_x\|_p < \varepsilon/3$, and so

$$\|f - f_x\|_p \leq \|f - g\|_p + \|g - g_x\|_p + \|g_x - f_x\|_p < \varepsilon$$

if $x \in V$. Finally, $f_x - f_y = (f - f_{y-x})_x$, so that $\|f_x - f_y\|_p < \varepsilon$ if $y - x \in V$, and the proof is complete.

1. Haar Measure and Convolution (3)

Note that the same theorem (with the same proof) is true with $C_0(G)$ in place of $L^p(G)$, but that it is false for $L^\infty(G)$, unless G is discrete.

1.1.6. Convolutions. For any pair of Borel functions f and g on the LCA group G we define their convolution $f * g$ by the formula

$$(1) \quad (f * g)(x) = \int_G f(x - y)g(y)dy$$

provided that

$$(2) \quad \int_G |f(x - y)g(y)|dy < \infty.$$

Note that the integral (1) can also be written in the form

$$(3) \quad \int_G f_v(x)g(y)dy$$

so that $f * g$ may be regarded as a limit of linear combinations of translates of f ; this statement may be made precise, but we assign it only heuristic value at present. (See Theorem 7.1.2.)

THEOREM. (a) If (2) holds for some $x \in G$, then $(f * g)(x) = (g * f)(x)$.

(b) If $f \in L^1(G)$ and $g \in L^\infty(G)$, then $f * g$ is bounded and uniformly continuous.

(c) If f and g are in $C_c(G)$, with compact supports A and B , then the support of $f * g$ lies in $A + B$, so that $f * g \in C_c(G)$.

(d) If $1 < p < \infty$, $1/p + 1/q = 1$, $f \in L^p(G)$, and $g \in L^q(G)$, then $f * g \in C_0(G)$.

(e) If f and g are in $L^1(G)$, then (2) holds for almost all $x \in G$, $f * g \in L^1(G)$, and the inequality

$$\|f * g\|_1 \leq \|f\|_1 \|g\|_1$$

holds.

(f) If f, g, h are in $L^1(G)$, then $(f * g) * h = f * (g * h)$.

Proof: Replacing y by $y + x$ in (1) and applying 1.1.4, we obtain

$$(f * g)(x) = \int_G f(-y)g(y+x)dy = \int_G f(y)g(-y+x)dy = (g * f)(x),$$

and (a) is proved.

Under the hypotheses of (b), it is clear that

$$|(f * g)(x)| \leq \|f\|_1 \|g\|_\infty \quad (x \in G)$$

so that $f * g$ is bounded. For $x \in G, z \in G$, we have

1. Haar Measure and Convolution (4)

$$\begin{aligned} |(f * g)(x) - (f * g)(z)| &\leq \int_G |f(x - y) - f(z - y)| |g(y)| dy \\ &\leq \|f_{-x} - f_{-z}\|_1 \|g\|_\infty. \end{aligned}$$

Theorem 1.1.5 shows that the last expression can be made arbitrarily small by restricting $x - z$ to lie in a suitably chosen neighborhood of 0 and (b) follows.

If f vanishes outside A and g vanishes outside B , then $f(x - y)g(y) = 0$ unless $y \in B$ and $x - y \in A$, i.e., unless $x \in A + B$. Thus $f * g$ vanishes outside $A + B$, and (c) is proved.

To prove (d), choose sequences $\{f_n\}$ and $\{g_n\}$ in $C_c(G)$ such that $\|f_n - f\|_p \rightarrow 0$ and $\|g_n - g\|_q \rightarrow 0$ as $n \rightarrow \infty$. Hölder's inequality shows that $f_n * g_n \rightarrow f * g$ uniformly. By (c), $f_n * g_n \in C_c(G)$. Hence $f * g \in C_0(G)$, and (d) follows.

The proof of (e) will depend on Fubini's theorem, and we first have to show that the integrand in (1) is a Borel function on $G \times G$. Fix an open set V in the plane, put $E = f^{-1}(V)$, $E' = E \times G$, and let $E'' = \{(x, y) : x - y \in E\}$. Then E' is a Borel set in $G \times G$, and since the homeomorphism of $G \times G$ onto itself which carries (x, y) to $(x + y, y)$ maps E' onto E'' , E'' is also a Borel set. Since $f(x - y) \in V$ if and only if $(x, y) \in E''$, we see that $f(x - y)$ is a Borel function on $G \times G$, and so is the product $f(x - y)g(y)$.

By Fubini's theorem,

$$\int_G \int_G |f(x - y)g(y)| dx dy = \|f\|_1 \|g\|_1.$$

Setting $\phi(x) = \int_G |f(x - y)g(y)| dy$, it follows that $\phi \in L^1(G)$. In particular, $\phi(x) < \infty$ for almost all x , and so $(f * g)(x)$ exists for almost all x . Finally, $|(f * g)(x)| \leq \phi(x)$, and the proof of (e) is complete.

The proof of (f) is also an application of Fubini's theorem, justified by (e) for almost all x :

$$\begin{aligned} (f * (g * h))(x) &= \int_G f(x - z)(g * h)(z) dz \\ &= \int_G \int_G f(x - z)g(z - y)h(y) dy dz \\ &= \int_G \int_G f(x - z - y)g(z)h(y) dy dz \\ &= \int_G (f * g)(x - y)h(y) dy = ((f * g) * h)(x). \end{aligned}$$

1. Haar Measure and Convolution (5)

1.1.7. THEOREM. *For any LCA group G , $L^1(G)$ is a commutative Banach algebra, if multiplication is defined by convolution. If G is discrete, $L^1(G)$ has a unit.*

Proof: The first statement follows from parts (e), (f), and (a) of Theorem 1.1.6, since the distributive law holds: $f * (g + h) = f * g + f * h$.

If G is discrete and the Haar measure is normalized as indicated in Section 1.1.3, then

$$(f * g)(x) = \sum_{y \in G} f(x - y)g(y),$$

and if $e(0) = 1$ but $e(x) = 0$ for all $x \neq 0$, then $e \in L^1(G)$ and $f * e = f$. Thus e is the unit of $L^1(G)$.

1.1.8. If G is not discrete, then $L^1(G)$ has no unit (see Section 1.7.3), but *approximate units* are always available.

THEOREM. *Given $f \in L^1(G)$ and $\varepsilon > 0$, there exists a neighborhood V of 0 in G with the following property: if u is a non-negative Borel function which vanishes outside V , and if $\int_G u(x)dx = 1$, then*

$$\|f - f * u\|_1 < \varepsilon.$$

Proof: By Theorem 1.1.5, we can choose V so that $\|f - f_y\|_1 < \varepsilon$ for all $y \in V$. If u satisfies the hypotheses, we have

$$(f * u)(x) - f(x) = \int_G [f(x - y) - f(x)]u(y)dy$$

so that

$$\begin{aligned} \|f * u - f\|_1 &\leq \int_G |u(y)|dy \int_G |f(x - y) - f(x)|dx \\ &= \int_V \|f - f_y\|_1 u(y)dy < \varepsilon. \end{aligned}$$

1. Haar Measure and Convolution (6)

1.2.1. Characters. A complex function γ on a LCA group G is called a *character* of G if $|\gamma(x)| = 1$ for all $x \in G$ and if the functional equation

$$(1) \quad \gamma(x + y) = \gamma(x)\gamma(y) \quad (x, y \in G)$$

is satisfied. The set of all *continuous* characters of G forms a group Γ , the *dual group* of G , if *addition* is defined by

$$(2) \quad (\gamma_1 + \gamma_2)(x) = \gamma_1(x)\gamma_2(x) \quad (x \in G; \gamma_1, \gamma_2 \in \Gamma).$$

Throughout this book, the letter Γ will denote the dual group of the LCA group G .

In view of the duality between G and Γ which will be established in Section 1.7, it is customary to write

$$(3) \quad (x, \gamma)$$

in place of $\gamma(x)$. With this notation, (1) and (2) become

$$(4) \quad (x + y, \gamma) = (x, \gamma)(y, \gamma) \text{ and } (x, \gamma_1 + \gamma_2) = (x, \gamma_1)(x, \gamma_2).$$

It follows immediately that

$$(5) \quad (0, \gamma) = (x, 0) = 1 \quad (x \in G, \gamma \in \Gamma)$$

and

$$(6) \quad (-x, \gamma) = (x, -\gamma) = (x, \gamma)^{-1} = \overline{(x, \gamma)}.$$

We shall presently endow Γ with a topology with respect to which Γ will itself be a LCA group. But first we identify Γ with the maximal ideal space of $L^1(G)$ (Appendix D).

1.2.2. THEOREM. *If $\gamma \in \Gamma$ and if*

$$(1) \quad \hat{f}(\gamma) = \int_G f(x)(-x, \gamma)dx \quad (f \in L^1(G)),$$

then the map $f \rightarrow \hat{f}(\gamma)$ is a complex homomorphism of $L^1(G)$, and is not identically 0. Conversely, every non-zero complex homomorphism of $L^1(G)$ is obtained in this way, and distinct characters induce distinct homomorphisms.

Proof: Suppose $f, g \in L^1(G)$, and $k = f * g$. Then

$$\begin{aligned} \hat{k}(\gamma) &= \int_G (f * g)(x)(-x, \gamma)dx = \int_G (-x, \gamma)dx \int_G f(x-y)g(y)dy \\ &= \int_G g(y)(-y, \gamma)dy \int_G f(x-y)(-x+y, \gamma)dx = \hat{g}(\gamma)\hat{f}(\gamma). \end{aligned}$$

Thus the map $f \rightarrow \hat{f}(\gamma)$ is multiplicative on the Banach algebra $L^1(G)$, and since it is clearly linear, it is a homomorphism. Since $|(-x, \gamma)| = 1$, $\hat{f}(\gamma) \neq 0$ for some $f \in L^1(G)$.

1. Haar Measure and Convolution (7)

For the converse, suppose h is a complex homomorphism of $L^1(G)$, $h \neq 0$. Then h is a bounded linear functional of norm 1 (Appendix D4), so that

$$(2) \quad h(f) = \int_G f(x) \phi(x) dx \quad (f \in L^1(G))$$

for some $\phi \in L^\infty(G)$ with $\|\phi\|_\infty = 1$ (Appendix E10). If f and g are in $L^1(G)$, we have

$$\begin{aligned} \int_G h(f)g(y)\phi(y)dy &= h(f)h(g) = h(f * g) = \int_G (f * g)(x)\phi(x)dx \\ &= \int_G g(y)dy \int_G f(x - y)\phi(x)dx = \int_G g(y)h(f_y)dy, \end{aligned}$$

so that

$$(3) \quad h(f)\phi(y) = h(f_y)$$

for almost all $y \in G$. By Theorem 1.1.5 and the continuity of h , the right side of (3) is a continuous function on G , for each $f \in L^1(G)$. Choosing f so that $h(f) \neq 0$, (3) shows that $\phi(y)$ coincides with a continuous function almost everywhere, and hence we may assume that ϕ is continuous, without affecting (2). Then (3) holds for all $y \in G$.

If we replace y by $x + y$ and then f by f_x in (3), we obtain $h(f)\phi(x + y) = h(f_{x+y}) = h((f_x)_y) = h(f_x)\phi(y) = h(f)\phi(x)\phi(y)$, so that

$$(4) \quad \phi(x + y) = \phi(x)\phi(y) \quad (x, y \in G).$$

Since $|\phi(x)| \leq 1$ for all x and since (4) implies that $\phi(-x) = \phi(x)^{-1}$, it follows that $|\phi(x)| = 1$. Hence $\phi \in \Gamma$.

Finally, if $\hat{f}(\gamma_1) = \hat{f}(\gamma_2)$ for all $f \in L^1(G)$, (1) implies that $(-x, \gamma_1) = (-x, \gamma_2)$ for almost all $x \in G$, and since γ_1 and γ_2 are continuous, 1.1.2 shows that the equality holds for all $x \in G$, so that $\gamma_1 = \gamma_2$.

2. The Dual Group and the Fourier-Transform (1)

1.2.3. The Fourier transform. For all $f \in L^1(G)$, the function $\hat{f}$ defined on Γ by

$$\hat{f}(\gamma) = \int_G f(x)(-x, \gamma) dx \quad (\gamma \in \Gamma)$$

is called the *Fourier transform* of f . The set of all functions $\hat{f}$ so obtained will be denoted throughout by $A(\Gamma)$.

By Theorem 1.2.2, $\hat{f}$ is precisely the Gelfand transform of f . If we give Γ the weak topology induced by $A(\Gamma)$ (Appendix A10), the basic facts of the Gelfand theory (Appendix D4) show that $A(\Gamma)$ is a separating subalgebra of $C_0(\Gamma)$. We summarize some of the properties of $A(\Gamma)$.

1.2.4. THEOREM. (a) $A(\Gamma)$ is a separating self-adjoint subalgebra of $C_0(\Gamma)$, so that $A(\Gamma)$ is dense in $C_0(\Gamma)$, by the Stone-Weierstrass theorem.

(b) The Fourier transform of $f * g$ is $\hat{f}\hat{g}$.

(c) $A(\Gamma)$ is invariant under translation and under multiplication by (x, γ) , for any $x \in G$.

(d) The Fourier transform, considered as a map of $L^1(G)$ into $C_0(\Gamma)$, is norm-decreasing and therefore continuous: $\|\hat{f}\|_\infty \leq \|f\|_1$.

(e) For $f \in L^1(G)$ and $\gamma \in \Gamma$, $(f * \gamma)(x) = (x, \gamma)\hat{f}(\gamma)$.

Proof: For $f \in L^1(G)$, define $\tilde{f}$ by

$$\tilde{f}(x) = \overline{f(-x)}.$$

The Fourier transform of $\tilde{f}$ is the complex conjugate of $\hat{f}$, and (a) follows; (b) is implicit in Theorem 1.2.2. If $\gamma_0 \in \Gamma$ and $g(x) = (x, \gamma_0)f(x)$, then $\hat{g}(\gamma) = \hat{f}(\gamma - \gamma_0)$, so that $A(\Gamma)$ is translation invariant. If $g = f_x$, then

$$\begin{aligned} \hat{g}(\gamma) &= \int_G f(y-x)(-y, \gamma) dy \\ &= (-x, \gamma) \int_G f(y-x)(x-y, \gamma) dy = (-x, \gamma)\hat{f}(\gamma). \end{aligned}$$

This proves (c); (d) and (e) are trivial; (e) allows us to interpret the Fourier transform as a convolution:

$$\hat{f}(\gamma) = (f * \gamma)(0) \quad (f \in L^1(G), \gamma \in \Gamma).$$

2. The Dual Group and the Fourier-Transform (2)

1.2.5. THEOREM. *If G is discrete, Γ is compact. If G is compact, Γ is discrete.*

Proof: If G is discrete, then $L^1(G)$ has a unit (Theorem 1.1.7) and its maximal ideal space Γ is therefore compact (Appendix D4).

If G is compact and its Haar measure is normalized so that $m(G) = 1$, the orthogonality relations

$$(1) \quad \int_G (x, \gamma) dx = \begin{cases} 1 & \text{if } \gamma = 0 \\ 0 & \text{if } \gamma \neq 0 \end{cases}$$

hold. The case $\gamma = 0$ is clear. If $\gamma \neq 0$, then $(x_0, \gamma) \neq 1$ for some $x_0 \in G$, and

$$\int_G (x, \gamma) dx = (x_0, \gamma) \int_G (x - x_0, \gamma) dx = (x_0, \gamma) \int_G (x, \gamma) dx,$$

so that (1) is proved. If $f(x) = 1$ for all $x \in G$, then $f \in L^1(G)$ since G is compact, and $\hat{f}(0) = 1$, $\hat{f}(\gamma) = 0$, if $\gamma \neq 0$, by (1). Since $\hat{f}$ is continuous, the set consisting of 0 alone is open in Γ , and so Γ is discrete.

1.2.6. The topology of Γ . So far, Γ is a group and a locally compact Hausdorff space. We shall now prove that these two structures fit together so as to make Γ a LCA group. Our proof depends on an alternative description of the topology of Γ :

THEOREM. (a) (x, γ) is a continuous function on $G \times \Gamma$.

(b) Let K and C be compact subsets of G and Γ , respectively, let U_r be the set of all complex numbers z with $|1 - z| < r$, and put

$$N(K, r) = \{\gamma: (x, \gamma) \in U_r \text{ for all } x \in K\},$$

$$N(C, r) = \{x: (x, \gamma) \in U_r \text{ for all } \gamma \in C\}.$$

Then $N(K, r)$ and $N(C, r)$ are open subsets of Γ and G , respectively.

(c) The family of all sets $N(K, r)$ and their translates is a base for the topology of Γ .

(d) Γ is a LCA group.

Proof: Equation (3) of Section 1.2.2, rewritten in the form

$$(1) \quad \hat{f}(\gamma)(x, \gamma) = \hat{f}_x(\gamma) \quad (x \in G, \gamma \in \Gamma)$$

implies (a), as soon as it is proved that $\hat{f}_x(\gamma)$ is a continuous function on $G \times \Gamma$, for every $f \in L^1(G)$.

2. The Dual Group and the Fourier-Transform (3)

Fix x_0, γ_0 , and $\varepsilon > 0$. There are neighborhoods V of x_0 and W of γ_0 such that

$$(2) \quad \|f_x - f_{x_0}\|_1 < \varepsilon \text{ and } |\hat{f}_{x_0}(\gamma) - \hat{f}_{x_0}(\gamma_0)| < \varepsilon$$

for all $x \in V, \gamma \in W$, by Theorem 1.1.5 and the continuity of $\hat{f}_{x_0}$. Since $|\hat{f}_x(\gamma) - \hat{f}_{x_0}(\gamma)| \leq \|f_x - f_{x_0}\|_1$, it follows that $|\hat{f}_x(\gamma) - \hat{f}_{x_0}(\gamma_0)| < 2\varepsilon$ if $x \in V$ and $\gamma \in W$, and (a) is proved.

Choose a compact set K in G , choose $r > 0$, and fix $\gamma_0 \in N(K, r)$. To every $x_0 \in K$ there correspond neighborhoods V of x_0 and W of γ_0 such that $(x, \gamma) \in U_r$, if $x \in V$ and $\gamma \in W$; this follows from (a). Since K is compact, finitely many of these sets V cover K , and if W^* is the intersection of the corresponding sets W , then $W^* \subset N(K, r)$. Since W^* is a neighborhood of γ_0 , $N(K, r)$ is open.

The same proof applies to $N(C, r)$.

To prove (c), assume that V is a neighborhood of γ_0 . We have to show that $\gamma_0 + N(K, r) \subset V$ for some choice of K and r . Take $\gamma_0 = 0$, without loss of generality. The definition of the Gelfand topology on Γ shows that there exist functions $f_1, \dots, f_n \in L^1(G)$ and $\varepsilon > 0$ so that

$$(3) \quad \bigcap_{i=1}^n \{\gamma: |\hat{f}_i(\gamma) - \hat{f}_i(0)| < \varepsilon\} \subset V.$$

Since $C_c(G)$ is dense in $L^1(G)$, we may assume that $f_1, \dots, f_n$ vanish outside a compact set K in G . If

$$(4) \quad r < \varepsilon / \max_i \|f_i\|_1$$

and if $\gamma \in N(K, r)$, then

$$(5) \quad |\hat{f}_i(\gamma) - \hat{f}_i(0)| \leq \int_K |(-x, \gamma) - 1| |f_i(x)| dx \leq r \|f_i\|_1 < \varepsilon.$$

Hence $N(K, r) \subset V$, and (c) follows.

Given $\gamma', \gamma'' \in \Gamma$ and $N(K, r)$, the obvious relation

$$(6) \quad [\gamma' + N(K, r/2)] - [\gamma'' + N(K, r/2)] \subset \gamma' - \gamma'' + N(K, r)$$

shows, by (b) and (c), that the map $(\gamma', \gamma'') \rightarrow \gamma' - \gamma''$ of $\Gamma \times \Gamma$ onto Γ is continuous. This completes the theorem.

2. The Dual Group and the Fourier-Transform (4)

1.2.7. EXAMPLES. The “classical groups” of Fourier analysis are:

- (a) the additive group R of the real numbers, with the natural topology of the real line;
- (b) the additive group of the reals modulo 2π , or, equivalently, the circle group T , the multiplicative group of all complex numbers of absolute value 1;
- (c) the additive group Z of the integers.

The circle group is of particular importance to us, since characters are nothing but homomorphisms into T .

Suppose $G = R$ and fix $\gamma \in \Gamma$. Write $\gamma(x)$ instead of (x, γ) , for the moment; there exists $\delta > 0$ such that

$$(1) \quad \int_0^\delta \gamma(t) dt = \alpha \neq 0.$$

The functional equation

$$(2) \quad \gamma(x+t) = \gamma(x)\gamma(t) \quad (x, t \in R)$$

then implies that

$$(3) \quad \alpha \cdot \gamma(x) = \gamma(x) \int_0^\delta \gamma(t) dt = \int_0^\delta \gamma(x+t) dt = \int_x^{x+\delta} \gamma(t) dt.$$

Since γ is continuous, the last expression is differentiable, and so γ has a continuous derivative γ' . Differentiate (2) with respect to t and then set $t = 0$. The result is the differential equation

$$(4) \quad \gamma'(x) = A\gamma(x), \quad A = \gamma'(0).$$

Since $\gamma(0) = 1$ and since γ is bounded, (4) implies that

$$(5) \quad \gamma(x) = e^{iyx}$$

for some $y \in R$. The correspondence $\gamma \leftrightarrow y$ is an isomorphism between Γ and R . Thus: *The dual group of R is R .*

We still have to check that the natural topology of R is the same as the Gelfand topology of the dual group. For $r > 0$ and $n = 1, 2, 3, \dots$, let $V(n, r)$ be the set of all y such that $|1 - e^{ixy}| < r$ if $|x| \leq n$. By Theorem 1.2.6, the sets $V(n, r)$ form a neighborhood base at 0 with respect to the Gelfand topology. But $y \in V(n, r)$ if and only if $|y| < (2/n) \arcsin(r/2)$. Thus the two topologies coincide.

If $G = T$, the same computation as above shows that every character of T must be of the form (5), but now we also must have $\gamma(x + 2\pi) = \gamma(x)$. Hence y must be an integer, and Γ is identified as the discrete group Z (compare Theorem 1.2.5).

2. The Dual Group and the Fourier-Transform (5)

If $G = Z$ and $\gamma \in T$, then $(1, \gamma) = e^{i\alpha}$ for some real α , and it follows that $(n, \gamma) = e^{in\alpha}$. The correspondence $\gamma \leftrightarrow e^{i\alpha}$ is an isomorphism between T and T , and we conclude that T is the dual group of Z (the two topologies coincide, as in the case $G = R$).

The Fourier transforms, in these three cases, have the following forms:

$$G = R: \quad \hat{f}(y) = \int_{-\infty}^{\infty} f(x) e^{-iyx} dx \quad (y \in R),$$

$$G = T: \quad \hat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(e^{i\theta}) e^{-in\theta} d\theta \quad (n \in Z),$$

$$G = Z: \quad \hat{f}(e^{i\alpha}) = \sum_{n=-\infty}^{\infty} f(n) e^{-in\alpha} \quad (e^{i\alpha} \in T).$$

3. Fourier Stieltjes Transforms (1)

1.3.1. Convolutions of measures. Suppose G is a LCA group, and μ, λ are members of $M(G)$ (Appendix E1), i.e., bounded regular complex valued measures on G . Let $\mu \times \lambda$ be their product measure on the product space $G^2 = G \times G$, and associate with each Borel set E in G the set

$$(1) \quad E_{(2)} = \{(x, y) \in G^2 : x + y \in E\}.$$

Then $E_{(2)}$ is a Borel set in G^2 (see the proof of Theorem 1.1.6(d)) and we define $\mu * \lambda$ by

$$(2) \quad (\mu * \lambda)(E) = (\mu \times \lambda)(E_{(2)}).$$

The set function $\mu * \lambda$ so defined is called the *convolution* of μ and λ .

1.3.2. THEOREM. (a) *If $\mu \in M(G)$ and $\lambda \in M(G)$, then $\mu * \lambda \in M(G)$.*

(b) *Convolution is commutative and associative.*

(c) $\|\mu * \lambda\| \leq \|\mu\| \cdot \|\lambda\|$.

COROLLARY. *$M(G)$ is a commutative Banach algebra with unit, if multiplication is defined by convolution.*

Proof: The Jordan decomposition theorem shows that in the proof of (a) it is enough to consider non-negative measures only. Since $\mu \times \lambda$ is a measure on G^2 , it is clear that $(\mu * \lambda)(E) = \sum (\mu * \lambda)(E_i)$ if E is the union of the disjoint Borel sets E_i ($i = 1, 2, 3, \dots$). If E is a Borel set in G and if $\varepsilon > 0$, the regularity of $\mu \times \lambda$ shows that there is a compact set $K \subset E_{(2)}$ such that

$$(\mu \times \lambda)(K) > (\mu \times \lambda)(E_{(2)}) - \varepsilon.$$

If C is the image of K under the map $(x, y) \rightarrow x + y$, then C is a compact subset of E , $K \subset C_{(2)}$, and hence

$$(\mu * \lambda)(C) = (\mu \times \lambda)(C_{(2)}) \geq (\mu \times \lambda)(K) > (\mu * \lambda)(E) - \varepsilon.$$

This establishes one half of the requirement that $\mu * \lambda$ be regular. The other half follows by complementation, and (a) is proved. (This argument applies to more general situations; see Stromberg [1].)

Since G is commutative, the condition $x + y \in E$ is the same as the condition $y + x \in E$, and hence $\mu * \lambda = \lambda * \mu$.

The simplest way to prove associativity is to extend the definition of convolution to the case of n measures $\mu_1, \dots, \mu_n \in M(G)$: with each Borel set E in G associate the set

$$(1) \quad E_n = \{(x_1, \dots, x_n) \in G^n : x_1 + \dots + x_n \in E\},$$

3. Fourier Stieltjes Transforms (2)

and put

$$(2) \quad (\mu_1 * \mu_2 * \dots * \mu_n)(E) = (\mu_1 \times \mu_2 \times \dots \times \mu_n)(E_{(n)}),$$

where the measure on the right is the ordinary product measure on the product space G^n . Associativity now follows from Fubini's theorem, and (b) is proved.

Let χ_E be the characteristic function of the Borel set E in G . The definition of $(\mu * \lambda)(E)$ is equivalent to the equation

$$(3) \quad \int_G \chi_E d(\mu * \lambda) = \int_G \int_G \chi_E(x + y) d\mu(x) d\lambda(y).$$

Hence if f is a *simple function* (a finite linear combination of characteristic functions of Borel sets), we have

$$(4) \quad \int_G f d(\mu * \lambda) = \int_G \int_G f(x + y) d\mu(x) d\lambda(y),$$

and since every bounded Borel function is the uniform limit of a sequence of simple functions, (4) holds for every bounded Borel function f . (One could use (4) as the definition of $\mu * \lambda$.) If $|f(x)| \leq 1$ for all $x \in G$, then $|\int_G f(x + y) d\mu(x)| \leq \|\mu\|$ for all $y \in G$, and hence the right side of (4) does not exceed $\|\mu\| \cdot \|\lambda\|$. This proves part (c) of the theorem.

As to the Corollary, it only remains to be shown that $M(G)$ has a unit. Let δ_0 be the unit mass concentrated at the point $x = 0$; i.e., $\delta_0(E) = 1$ if $0 \in E$ and $\delta_0(E) = 0$ otherwise. Then $\mu * \delta_0 = \mu$ for all $\mu \in M(G)$, and the proof is complete.

1.3.3. Fourier-Stieltjes transforms. If $\mu \in M(G)$, the function $\hat{\mu}$ defined on Γ by

$$(1) \quad \hat{\mu}(\gamma) = \int_G (x, \gamma) d\mu(x) \quad (\gamma \in \Gamma)$$

is called the Fourier-Stieltjes transform of μ . The set of all such functions $\hat{\mu}$ will be denoted by $B(\Gamma)$.

THEOREM. (a) Each $\hat{\mu} \in B(\Gamma)$ is bounded and uniformly continuous.

(b) If $\sigma = \mu * \lambda$, then $\hat{\sigma} = \hat{\mu} \cdot \hat{\lambda}$. Hence the map $\mu \rightarrow \hat{\mu}(\gamma)$ is, for each $\gamma \in \Gamma$, a complex homomorphism of $M(G)$.

(c) $B(\Gamma)$ is invariant under translation, under multiplication by (x, γ) for any $x \in G$, and under complex conjugation.

Proof: The definition of $\hat{\mu}$ shows immediately that $|\hat{\mu}(\gamma)| \leq \|\mu\|$ for all $\gamma \in \Gamma$. Given $\delta > 0$, the regularity of $|\mu|$ shows that there is a compact set K in G such that $|\mu|(K') < \delta$, where K' is the complement of K . For any $\gamma_1, \gamma_2 \in \Gamma$ we have

$$|\hat{\mu}(\gamma_1) - \hat{\mu}(\gamma_2)| \leq \int_G |1 - (x, \gamma_1 - \gamma_2)| d|\mu|(x) = \int_K + \int_{K'}.$$

If $\gamma_1 - \gamma_2 \in N(K, \delta)$, as defined in Theorem 1.2.6, the above integrand is less than δ for $x \in K$, hence $\int_K$ does not exceed $\delta \|\mu\|$. The second integral is less than $2|\mu|(K') < 2\delta$. Hence $\hat{\mu}$ is uniformly continuous.

3. Fourier Stieltjes Transforms (3)

Suppose $\sigma = \mu * \lambda$. Formula (4) in the proof of Theorem 1.3.2 then implies that

$$\begin{aligned}\hat{\sigma}(\gamma) &= \int_G \langle -z, \gamma \rangle d(\mu * \lambda)(z) = \int_G \int_G \langle -x - y, \gamma \rangle d\mu(x) d\lambda(y) \\ &= \int_G \langle -x, \gamma \rangle d\mu(x) \int_G \langle -y, \gamma \rangle d\lambda(y) = \hat{\mu}(\gamma) \hat{\lambda}(\gamma),\end{aligned}$$

and (b) is proved.

The proof of (c) is quite similar to that of the analogous part of Theorem 1.2.4. If $d\lambda(x) = \langle x, \gamma_0 \rangle d\mu(x)$, then $\hat{\lambda}(\gamma) = \hat{\mu}(\gamma - \gamma_0)$. If $\lambda(E) = \mu(E - x)$, then $\hat{\lambda}(\gamma) = \langle x, \gamma \rangle \hat{\mu}(\gamma)$. If $\tilde{\mu}(E) = \mu(-E)$, then the Fourier-Stieltjes transform of $\tilde{\mu}$ is the complex conjugate of $\hat{\mu}$.

1.3.4. $L^1(G)$ as a subalgebra of $M(G)$. Every $f \in L^1(G)$ generates a measure $\mu_f \in M(G)$, defined by

$$(1) \quad \mu_f(E) = \int_E f(x) dx,$$

and which is absolutely continuous with respect to the Haar measure of G . Conversely, the Radon-Nikodym theorem (Appendix E9) shows that every absolutely continuous $\mu \in M(G)$ is μ_f for some $f \in L^1(G)$. Since we identify functions in $L^1(G)$ which differ only on a set of Haar measure 0, the correspondence between f and μ_f is one-to-one, and we may therefore regard $L^1(G)$ as a subset of $M(G)$. It is easily seen that $\hat{f}(\gamma) = \hat{\mu}_f(\gamma)$ for all $\gamma \in \Gamma$ and that

$\|f\|_1 = \|\mu_f\|$. Hence we may use f in place of μ_f without causing confusion. For instance, we may write $f * \sigma$ if $f \in L^1(G)$ and $\sigma \in M(G)$, instead of $\mu_f * \sigma$.

1.3.5. Let $M_c(G)$ and $M_d(G)$ denote the sets of all continuous and discrete members of $M(G)$, respectively (Appendix E6).

THEOREM. (a) $L^1(G)$ and $M_c(G)$ are closed ideals in $M(G)$.
(b) $M_d(G)$ is a closed subalgebra of $M(G)$.

Proof: If we apply the Fubini theorem to the definition of $\mu * \lambda$, we obtain, for any Borel set E in G ,

$$(1) \quad (\mu * \lambda)(E) = \int_G \mu(E - y) d\lambda(y).$$

If μ is absolutely continuous and $m(E) = 0$, then $m(E - y) = 0$ for all y , hence $\mu(E - y) = 0$, and so $(\mu * \lambda)(E) = 0$ for every $\lambda \in M(G)$. This says that $\mu * \lambda$ is absolutely continuous, and hence $L^1(G)$ is an ideal in $M(G)$. Since $\|f\|_1 = \|\mu_f\|$ and since $L^1(G)$ is complete, $L^1(G)$ is closed in $M(G)$. If E is countable, $\mu_n \in M_c(G)$, and $\|\mu - \mu_n\| \rightarrow 0$, then

$$|\mu(E)| = |(\mu - \mu_n)(E)| \leq \|\mu - \mu_n\|(E) \leq \|\mu - \mu_n\|,$$

so that $\mu(E) = 0$ and $\mu \in M_c(G)$. Thus $M_c(G)$ is closed, and part (a) is proved. Part (b) follows from the observation that the convolution of two point-measures is a point-measure.

3. Fourier Stieltjes Transforms (4)

1.3.6. A uniqueness theorem. We shall see later that $\hat{\mu}$ determines μ , i.e. if $\mu \in M(G)$ and $\hat{\mu} = 0$, then $\mu = 0$. At present, we can prove this for the inverse transform:

THEOREM. *If $\mu \in M(\Gamma)$ and if*

$$\int_{\Gamma} (x, \gamma) d\mu(\gamma) = 0$$

for every $x \in G$, then $\mu = 0$.

Proof: For every $f \in L^1(G)$,

$$\begin{aligned} \int_{\Gamma} \hat{f}(\gamma) d\mu(\gamma) &= \int_{\Gamma} \int_G f(x) (-x, \gamma) dx d\mu(\gamma) \\ &= \int_G f(x) dx \int_{\Gamma} (-x, \gamma) d\mu(\gamma) = 0. \end{aligned}$$

Since $A(\Gamma)$ is dense in $C_0(\Gamma)$ (Theorem 1.2.4), it follows that $\int_{\Gamma} \phi d\mu = 0$ for every $\phi \in C_0(\Gamma)$, and hence $\mu = 0$.

4. Positive Definite Functions (1)

1.4.1. A function ϕ , defined on G , is said to be *positive-definite* if the inequality

$$(1) \quad \sum_{n,m=1}^N c_n \overline{c_m} \phi(x_n - x_m) \geq 0$$

holds for every choice of $x_1, \dots, x_N$ in G and for every choice of complex numbers $c_1, \dots, c_N$.

If ϕ is positive-definite, the following three relations hold:

$$(2) \quad \phi(-x) = \overline{\phi(x)};$$

$$(3) \quad |\phi(x)| \leq \phi(0);$$

$$(4) \quad |\phi(x) - \phi(y)|^2 \leq 2\phi(0) \operatorname{Re} [\phi(0) - \phi(x - y)].$$

We conclude from (3) that $\phi(0) \geq 0$ and that ϕ is bounded; (4) implies that ϕ is uniformly continuous if ϕ is continuous at 0.

To prove these relations, take $N = 2$ in (1); $x_1 = 0$, $x_2 = x$; $c_1 = 1$, $c_2 = c$. This gives

$$(5) \quad \{1 + |c|^2\}\phi(0) + c\phi(x) + \bar{c}\phi(-x) \geq 0.$$

Taking $c = 1$, we see that $\phi(x) + \phi(-x)$ is real; $c = i$ shows that $i(\phi(x) - \phi(-x))$ is real. Hence (2) holds.

If c is chosen so that $c\phi(x) = -|\phi(x)|$, (5) implies (3). To prove (4), take $N = 3$ in (1); $x_1 = 0$, $x_2 = x$, $x_3 = y$; $c_1 = 1$, λ real,

$$c_2 = \frac{\lambda|\phi(x) - \phi(y)|}{\phi(x) - \phi(y)},$$

and $c_3 = -c_2$. Then (1) simplifies to

$$(6) \quad \phi(0)(1 + 2\lambda^2) + 2\lambda|\phi(x) - \phi(y)| - 2\lambda^2 \operatorname{Re} \phi(x - y) \geq 0.$$

The discriminant of the quadratic polynomial (6) in λ can therefore not be positive, and this gives (4).

1.4.2. Examples of positive-definite functions. (a) Suppose $f \in L^2(G)$ and $\phi = f * \bar{f}$. Then ϕ is positive-definite and continuous on G .

The convolution of any two functions in $L^2(G)$ is continuous (Theorem 1.1.6(d)) and

$$\begin{aligned} \sum c_n \overline{c_m} \phi(x_n - x_m) &= \sum c_n \overline{c_m} \int_G f(x_n - x_m - y) \overline{f(-y)} dy \\ &= \sum c_n \overline{c_m} \int_G f(x_n - y) \overline{f(x_m - y)} dy = \int_G |\sum c_n f(x_n - y)|^2 dy \geq 0. \end{aligned}$$

4. Positive Definite Functions (2)

(b) Every character is positive-definite, hence so is every finite linear combination of characters if the coefficients are positive. More generally, if $\mu \in M(\Gamma)$, if $\mu \geq 0$, and if

$$(1) \quad \phi(x) = \int_{\Gamma} (x, \gamma) d\mu(\gamma) \quad (x \in G),$$

then ϕ is continuous and positive definite.

Indeed, (1) shows that

$$\begin{aligned} \sum c_n \overline{c_m} \phi(x_n - x_m) &= \int_{\Gamma} \sum_{n,m} c_n \overline{c_m} (x_n - x_m, \gamma) d\mu(\gamma) \\ &= \int_{\Gamma} \left| \sum_n c_n (x_n, \gamma) \right|^2 d\mu(\gamma) \geq 0, \end{aligned}$$

so that ϕ is positive-definite. Since the sets $N(C, r)$ of Theorem 1.2.6 are open in G , our proof of the continuity of $\hat{\mu}$ (Theorem 1.3.3) shows equally well that ϕ is continuous if ϕ is defined by (1).

1.4.3. The previous example (1.4.2(b)) establishes the trivial half of the following important characterization of positive-definite functions:

BOCHNER'S THEOREM. *A continuous function ϕ on G is positive-definite if and only if there is a non-negative measure $\mu \in M(\Gamma)$ such that*

$$(1) \quad \phi(x) = \int_{\Gamma} (x, \gamma) d\mu(\gamma) \quad (x \in G).$$

For $G = Z$, this is due to Herglotz [1]; for $G = R$, to Bochner [1]; for the general case, to Weil [1]. Bochner was the first to recognize the key role which this result plays in harmonic analysis. By 1.3.6, the above representation (1) is unique.

Proof: Suppose ϕ is continuous and positive-definite. By 1.4.1(3) we may assume, without loss of generality, that $\phi(0) = 1$.

If $f \in C_c(G)$ and has support K , then $f(x)\overline{f(y)}\phi(x - y)$ is uniformly continuous on $K \times K$, and K can be partitioned into disjoint sets $E_1, \dots, E_n$ such that the sum

$$(2) \quad \sum_{i,j=1}^n f(x_i)\overline{f(x_j)}\phi(x_i - x_j)m(E_i)m(E_j) \quad (x_i \in E_i)$$

4. Positive Definite Functions (3)

differs from the integral

$$(3) \quad \int_G \int_G f(x) \overline{f(y)} \phi(x-y) dx dy$$

by as little as we please. Since ϕ is positive-definite, (2) is always non-negative, and hence so is (3). Since $C_c(G)$ is dense in $L^1(G)$, it follows that (3) is non-negative for every $f \in L^1(G)$.

Define a functional T_ϕ by

$$(4) \quad T_\phi(f) = \int_G f(x) \phi(x) dx \quad (f \in L^1(G))$$

and put

$$(5) \quad [f, g] = T_\phi(f * \tilde{g}) \quad (f, g \in L^1(G)).$$

We recall that $\tilde{g}(x) = \overline{g(-x)}$, so that

$$(6) \quad [f, g] = \int_G \int_G f(x) \overline{g(y)} \phi(x-y) dx dy.$$

Hence $[f, g]$ is linear in f , $[g, f]$ is the complex conjugate of $[f, g]$, and $[f, f] \geq 0$. These are just the properties of the Hilbert space inner product which are needed for the standard proof of the Schwarz inequality. In our case, the inequality is

$$(7) \quad |[f, g]|^2 \leq [f, f][g, g].$$

Take for g the characteristic function of a symmetric neighborhood V of 0, divided by $m(V)$. By (6),

$$[f, g] - T_\phi(f) = \int_G f(x) \frac{1}{m(V)} \int_V [\phi(x-y) - \phi(x)] dy dx$$

and

$$[g, g] - 1 = \frac{1}{m(V)^2} \int_V \int_V [\phi(x-y) - 1] dx dy.$$

Since ϕ is uniformly continuous, these expressions can be made arbitrarily small by taking V small enough, and then (7) yields the inequality

$$(8) \quad |T_\phi(f)|^2 \leq [f, f] = T_\phi(f * \tilde{f}) \quad (f \in L^1(G)).$$

Put $h = f * \tilde{f}$ and $h^n = h^{n-1} * h$ ($n = 2, 3, 4, \dots$). Since $\|\phi\|_\infty = 1$, we have $\|T_\phi\| = 1$, and if we apply (8) with $h, h^2, h^4, \dots$ in place of f , we obtain

$$|T_\phi(f)|^2 \leq T_\phi(h) \leq \{T_\phi(h^2)\}^{\frac{1}{2}} \leq \dots \leq \{T_\phi(h^{2^n})\}^{2^{-n}} \leq \|h^{2^n}\|_1^{2^{-n}}.$$

As $n \rightarrow \infty$, the last expression converges to the spectral radius of h , i.e. to $\|\hat{h}\|_\infty$. (See Appendix D 6 and Theorem 1.2.2.) Hence

$$(9) \quad |T_\phi(f)|^2 \leq \|\hat{h}\|_\infty = \|\hat{f}\|_\infty^2, \text{ or } |T_\phi(f)| \leq \|\hat{f}\|_\infty \quad (f \in L^1(G)).$$

4. Positive Definite Functions (4)

This means that T_ϕ may be regarded as a bounded linear functional on $A(I)$, with respect to the supremum norm. (We have not yet proved that $\hat{f}_1 = \hat{f}_2$ implies $f_1 = f_2$, but (9) shows that $\hat{f}_1 = \hat{f}_2$ implies $T_\phi(f_1) = T_\phi(f_2)$, and this is sufficient.) We can extend T_ϕ to a bounded linear functional on $C_0(I)$, preserving its norm, and the Riesz representation theorem then implies that there is a $\mu \in M(I)$, with $\|\mu\| \leq 1$, such that

$$(10) \quad T_\phi(f) = \int_I \hat{f}(-\gamma) d\mu(\gamma) = \int_G f(x) dx \int_I (x, \gamma) d\mu(\gamma).$$

Comparison of (10) and (4) shows that (1) holds for almost all $x \in G$, hence for all x , since both sides of (1) are continuous. Finally, taking $x = 0$ in (1), we have

$$1 = \phi(0) = \int_I d\mu(\gamma) = \mu(I) \leq \|\mu\| = 1;$$

hence $\mu(I) = \|\mu\|$, and this implies that $\mu \geq 0$.

5. The Inversion Theorem (1)

1.5.1. We let $B(G)$ be the set of all functions f on G which are representable in the form

$$(1) \quad f(x) = \int_G (x, \gamma) d\mu(\gamma) \quad (x \in G).$$

Bochner's theorem implies, in combination with the Jordan decomposition theorem, that $B(G)$ is exactly the set of all finite linear combinations of continuous positive-definite functions on G .

THEOREM. (a) If $f \in L^1(G) \cap B(G)$, then $\hat{f} \in L^1(\Gamma)$.

(b) If the Haar measure of G is fixed, the Haar measure of Γ can be so normalized that the inversion formula

$$(2) \quad f(x) = \int_\Gamma \hat{f}(\gamma) (x, \gamma) d\gamma \quad (x \in G)$$

is valid for every $f \in L^1(G) \cap B(G)$.

Proof: Let us write B^1 in place of $L^1(G) \cap B(G)$, and if μ is associated with f as in (1) above, let us write $\mu = \mu_f$. (This notation has nothing to do with our earlier use of the symbol μ_f in Section 1.3.4.) If $f \in B^1$ and $h \in L^1(G)$, we then have

$$(3) \quad (h * f)(0) = \int_G h(-x) f(x) dx = \int_\Gamma \hat{h}(\gamma) d\mu_f(\gamma),$$

and if g is also in B^1 , (3) implies that

$$\int_\Gamma \hat{h} \hat{g} d\mu_f = ((h * g) * f)(0) = ((h * f) * g)(0) = \int_\Gamma \hat{h} \hat{f} d\mu_g.$$

Since $A(\Gamma)$ is dense in $C_0(\Gamma)$, it follows that

$$(4) \quad \hat{g} d\mu_f = \hat{f} d\mu_g. \quad (f, g \in B^1).$$

We shall now define a positive linear functional T on $C_c(\Gamma)$. Suppose K is the support of some $\psi \in C_c(\Gamma)$. To every $\gamma_0 \in K$ there corresponds a function $u \in C_c(G)$ with $\hat{u}(\gamma_0) \neq 0$, since $C_c(G)$ is dense in $L^1(G)$. The Fourier transform of $u * \tilde{u}$ is positive at γ_0 , and is nowhere negative. Since K is compact, there is a finite number of such functions, say $u_1, \dots, u_n$ such that the function $g = u_1 * \tilde{u}_1 + \dots + u_n * \tilde{u}_n$ has $\hat{g} > 0$ on K . Since $g \in C_c(G)$, 1.4.2(a) shows that $g \in B^1$. Put

$$(5) \quad T\psi = \int_\Gamma (\psi/\hat{g}) d\mu_g.$$

Note that $T\psi$ is well defined: if g were replaced by another function f in B^1 whose Fourier transform has no zero on K , the value of $T\psi$ would not be changed, since (4) implies that

$$(6) \quad \int \frac{\psi}{\hat{f} \hat{g}} \hat{f} d\mu_g = \int \frac{\psi}{\hat{f} \hat{g}} \hat{g} d\mu_f.$$

5. The Inversion Theorem (2)

It is clear that T is linear. The function g in (5) is positive-definite, hence $\mu_g \geq 0$, and it follows that $T\psi \geq 0$ if $\psi \geq 0$. There exists ψ and μ_f such that $\int \psi d\mu_f \neq 0$, and if g is as in (5), we have

$$(7) \quad T(\psi\hat{f}) = \int_{\Gamma} (\psi\hat{f}/\hat{g}) d\mu_g = \int_{\Gamma} \psi d\mu_f \neq 0.$$

Thus $T \neq 0$.

Fix $\psi \in C_c(\Gamma)$ and $\gamma_0 \in \Gamma$. Construct g as above, so that $\hat{g} > 0$ on K and also on $K + \gamma_0$. Setting $f(x) = (-x, \gamma_0)g(x)$, we have $\hat{f}(\gamma) = \hat{g}(\gamma + \gamma_0)$ and $\mu_f(E) = \mu_g(E - \gamma_0)$. If $\psi_0(\gamma) = \psi(\gamma - \gamma_0)$, then

$$T\psi_0 = \int_{\Gamma} [\psi(\gamma - \gamma_0)/\hat{g}(\gamma)] d\mu_g(\gamma) = \int_{\Gamma} [\psi(\gamma)/\hat{f}(\gamma)] d\mu_f(\gamma) = T\psi.$$

Thus T is translation-invariant, and it follows that

$$(8) \quad T\psi = \int_{\Gamma} \psi(\gamma) d\gamma \quad (\gamma \in C_c(\Gamma)),$$

where $d\gamma$ denotes a Haar measure on Γ .

If now $f \in B^1$ and $\psi \in C_c(\Gamma)$, (7) and (8) show that

$$(9) \quad \int_{\Gamma} \psi d\mu_f = T(\psi\hat{f}) = \int_{\Gamma} \psi\hat{f} d\gamma,$$

and since (9) holds for every $\psi \in C_c(\Gamma)$, we conclude that

$$(10) \quad \hat{f} d\gamma = d\mu_f \quad (f \in B^1).$$

Since μ_f is a finite measure, it follows that $\hat{f} \in L^1(\Gamma)$, and substitution of (10) into (1) gives the inversion formula (2).

This completes the proof.

1.5.2. Consequences of the inversion theorem. Let V be a neighborhood of 0 in G , choose a compact neighborhood W of 0 such that $W - W \subset V$, let f be the characteristic function of W , divided by $m(W)^{\frac{1}{2}}$, and put $g = f * \bar{f}$. Then g is continuous, positive-definite (by 1.4.2(a)), and 0 outside $W - W$. The inversion theorem therefore applies to g . Hence $\hat{g} = |\hat{f}|^2 \geq 0$,

$$(1) \quad \int_{\Gamma} \hat{g}(\gamma) d\gamma = g(0) = 1,$$

and it follows that there is a compact set C in Γ such that

$$(2) \quad \int_C \hat{g}(\gamma) d\gamma > \frac{2}{3}.$$

If $x \in N(C, 1/3)$ (in the notation of Theorem 1.2.6), we write

$$(3) \quad g(x) = \left(\int_C + \int_{C'} \right) \hat{g}(\gamma)(x, \gamma) d\gamma;$$

for $\gamma \in C$, $|1 - (x, \gamma)| < 1/3$, hence $\operatorname{Re}(x, \gamma) > 2/3$, and the integral over C is at least $2/3 \int_C \hat{g} > 4/9$. Since $|\int_{C'}| < 1/3$, we see that $g(x) > 1/9$ if $x \in N(C, 1/3)$, and our conclusion is: $N(C, 1/3) \subset V$.

5. The Inversion Theorem (3)

Since the sets $N(C, r)$ are open in G (Theorem 1.2.6(b)), we now have the following analogue of 1.2.6(c):

The family of all sets $N(C, r)$ and their translates is a base for the topology of G .

If $x_0 \in G$, $x_0 \neq 0$, we can choose V in the preceding paragraph so that $x_0 \notin V$, and we conclude that $(x_0, \gamma) \neq 1$ for some $\gamma \in \Gamma$. Hence Γ separates points on G : If $x_1 \neq x_2$, then $(x_1 - x_2, \gamma) \neq 1$ for some γ , and so $(x_1, \gamma) \neq (x_2, \gamma)$.

Any function of the form

$$f(x) = \sum_{j=1}^n a_j(x, \gamma_j) \quad (x \in G)$$

is called a *trigonometric polynomial* on G . The set of all trigonometric polynomials on G is an algebra over the complex field, with respect to pointwise multiplication, and is closed under complex conjugation. Since Γ separates points on G , the Stone-Weierstrass theorem yields the following result:

If G is compact, the trigonometric polynomials on G form a dense subalgebra of $C(G)$.

It follows that the trigonometric polynomials are also dense in $L^p(G)$, $1 \leq p < \infty$, if G is compact (see Appendix E8).

1.5.3. Normalization of Haar measure. If the Haar measure of G is given, the inversion theorem singles out a specific Haar measure of Γ , adjusted so that the inversion formula holds. In Section 1.1.3 we introduced standard normalizations for the Haar measures of compact and discrete groups. Since Γ is compact [discrete] if G is discrete [compact] (Theorem 1.2.5) the question arises whether these normalizations are “correct,” i.e., whether the inversion formula holds for them.

To prove that this is so, it suffices to consider just one function (not identically 0) and its Fourier transform.

If G is compact and $m(G) = 1$, take $f(x) = 1$. Then (see 1.2.5) $\hat{f}(0) = 1$ and $\hat{f}(\gamma) = 0$ if $\gamma \neq 0$. If m_Γ is the Haar measure of Γ , adjusted in accordance with the inversion theorem, then

$$(1) \quad 1 = f(0) = \int_\Gamma \hat{f}(\gamma) d\gamma = m_\Gamma(\{0\}),$$

and so m_Γ assigns measure 1 to each point of Γ .

5. The Inversion Theorem (4)

If G is discrete and each point has measure 1, take $f(0) = 1$, $f(x) = 0$ if $x \neq 0$. Then $\hat{f}(\gamma) = 1$, and

$$(2) \quad m(\Gamma) = \int_{\Gamma} \hat{f}(\gamma) d\gamma = f(0) = 1$$

if the inversion theorem holds.

To consider a non-trivial case, take $G = \mathbb{R}$ (see 1.2.7) so that $\Gamma = \mathbb{R}$, and let αdx , βdt be Haar measures on G and Γ ; here dx and dt denote ordinary Lebesgue measure on the real line. Since $e^{-|t|} > 0$, the easily verified formula

$$(3) \quad \frac{2\beta}{1+x^2} = \int_{-\infty}^{\infty} e^{-|t|} e^{ixt} \beta dt$$

shows that $(1+x^2)^{-1}$ is positive-definite, and the uniqueness of the inverse transform, combined with the inversion theorem, shows that

$$(4) \quad e^{-|t|} = 2\alpha\beta \int_{-\infty}^{\infty} \frac{e^{-ixt}}{1+x^2} dx.$$

With $t = 0$, (4) becomes

$$(5) \quad 1 = 2\alpha\beta \int_{-\infty}^{\infty} \frac{dx}{1+x^2} = 2\pi\alpha\beta,$$

and this is the normalization condition which α and β must satisfy.

Two of the possible choices are frequently used: $\alpha = 1/2\pi$, $\beta = 1$ or $\alpha = \beta = (2\pi)^{-1/2}$.

From now on, it will allways be tacitly assumed that the Haar measures of G and Γ are so adjusted that the inversion theorem holds.

6. The Plancherel Theorem (1)

1.6.1. THEOREM. *The Fourier transform, restricted to $(L^1 \cap L^2)(G)$, is an isometry (with respect to the L^2 -norms) onto a dense linear subspace of $L^2(\Gamma)$. Hence it may be extended, in a unique manner, to an isometry of $L^2(G)$ onto $L^2(\Gamma)$.*

Proof: If $f \in (L^1 \cap L^2)(G)$ and $g = f * \check{f}$, then $g \in L^1(G)$, g is continuous and positive definite, $|\hat{g}| = |\hat{f}|^2$, and the inversion theorem gives

$$\int_G |f(x)|^2 dx = \int_G f(x) \check{f}(-x) dx = g(0) = \int_\Gamma \hat{g}(\gamma) d\gamma = \int_\Gamma |\hat{f}(\gamma)|^2 d\gamma,$$

or $\|\hat{f}\|_2 = \|f\|_2$.

Let Φ be the set of all $\hat{f} \in A(\Gamma)$ with $f \in (L^1 \cap L^2)(G)$. Since $(L^1 \cap L^2)(G)$ is translation invariant, Φ is invariant under multiplication by (x, γ) , for any $x \in G$. Thus if $\psi \in L^2(\Gamma)$ and $\int_\Gamma \phi \bar{\psi} d\gamma = 0$ for all $\phi \in \Phi$, then also

$$\int_\Gamma \phi(\gamma) \overline{\psi(\gamma)}(x, \gamma) d\gamma = 0 \quad (\phi \in \Phi, x \in G).$$

Since $\phi \bar{\psi} \in L^1(\Gamma)$, the uniqueness theorem 1.3.6 implies that $\phi \bar{\psi} = 0$ almost everywhere, for every $\phi \in \Phi$. But $(L^1 \cap L^2)(G)$ is invariant under multiplication by (x, γ) , for any $\gamma \in \Gamma$, and so Φ is translation invariant. Hence to every γ_0 there corresponds a $\phi \in \Phi$ which is different from 0 in a neighborhood of γ_0 . It follows that $\psi = 0$ almost everywhere. Thus 0 is the only element of $L^2(\Gamma)$ which is orthogonal to Φ , and hence Φ is dense in $L^2(\Gamma)$ (see Appendix C12).

1.6.2. The above extension of the Fourier transform to $L^2(G)$ is sometimes referred to as the *Plancherel transform*; the symbol $\hat{f}$ will be used in this context as well. An important part of the theorem is the assertion that each function in $L^2(\Gamma)$ is the Plancherel transform of some $f \in L^2(G)$. For compact G this is a special case of the Riesz-Fischer theorem about orthogonal systems of functions (Zygmund [1], vol. I, p. 127).

If f and g are in $L^2(G)$, the identity

$$4f\bar{g} = |f + g|^2 - |f - g|^2 + i|f + ig|^2 - i|f - ig|^2,$$

combined with the isometric character of the Plancherel transform, yields the *Parseval formula*

$$\int_G f(x) \overline{g(x)} dx = \int_\Gamma \hat{f}(\gamma) \overline{\hat{g}(\gamma)} d\gamma.$$

6. The Plancherel Theorem (2)

1.6.3. THEOREM. $A(\Gamma)$ consists precisely of the convolutions $F_1 * F_2$, with F_1 and F_2 in $L^2(\Gamma)$.

Proof: Suppose $f, g \in L^2(G)$. Replacing g by $\tilde{g}$, the Parseval formula assumes the form

$$(1) \quad \int_G f(x)g(x)dx = \int_{\Gamma} \hat{f}(\gamma)\hat{g}(-\gamma)d\gamma,$$

and if we replace $g(x)$ by $(-x, \gamma_0)g(x)$ in (1), we obtain

$$(2) \quad \int_G f(x)g(x)(-x, \gamma_0)dx = \int_{\Gamma} \hat{f}(\gamma)\hat{g}(\gamma_0 - \gamma)d\gamma = (\hat{f} * \hat{g})(\gamma_0).$$

On the one hand, every $h \in L^1(G)$ is a product $h = fg$, with $f, g \in L^2(G)$, and (2) shows that $\hat{h} = \hat{f} * \hat{g}$, with $\hat{f}, \hat{g} \in L^2(\Gamma)$, by the Plancherel theorem. On the other hand, we can start with $\hat{f}, \hat{g} \in L^2(\Gamma)$, and see from (2) that $\hat{f} * \hat{g} \in A(\Gamma)$.

1.6.4. THEOREM. If E is a non-empty open set in Γ , there exists $\hat{f} \in A(\Gamma)$, $\hat{f} \neq 0$, such that $\hat{f}(\gamma) = 0$ outside E .

Proof: Let K be a compact subset of E , with $m(K) > 0$, let V be a compact neighborhood of 0 such that $K + V \subset E$, and set $\hat{f} = \hat{g} * \hat{h}$, where $\hat{g}$ and $\hat{h}$ are the characteristic functions of K and V , respectively. Then $\hat{f}(\gamma) = 0$ outside $K + V$, $\hat{f} \in A(\Gamma)$ by Theorem 1.6.3, and $\int_{\Gamma} \hat{f}(\gamma)d\gamma = m(K)m(V) > 0$, so that $\hat{f}$ is not identically 0.

7. The Pontryagin Duality Theorem (1)

1.7.1. If G is a LCA group, we have seen (Theorem 1.2.6) that its dual Γ is also a LCA group. Hence Γ has a dual group, say $\hat{\Gamma}$, and everything we have proved so far for the ordered pair (G, Γ) holds equally well for the pair $(\Gamma, \hat{\Gamma})$. The value of a character $\hat{\gamma} \in \hat{\Gamma}$ at the point $\gamma \in \Gamma$ will be written $(\gamma, \hat{\gamma})$. (This notation is temporary, and will be abandoned as soon as we prove that $\hat{\Gamma} = G$.)

By Theorem 1.2.6(a) every $x \in G$ may be regarded as a continuous character on Γ , and thus there is a natural map α of G into $\hat{\Gamma}$, defined by

$$(1) \quad (x, \gamma) = (\gamma, \alpha(x)) \quad (x \in G, \gamma \in \Gamma).$$

1.7.2. THEOREM. *The above map α is an isomorphism and a homeomorphism of G onto $\hat{\Gamma}$.*

Thus $\hat{\Gamma}$ may be identified with G , and a more informal statement of the result would be:

Every LCA group is the dual group of its dual group.

This is the Pontryagin duality theorem.

Proof: For $x, y \in G$ and $\gamma \in \Gamma$, we have

$$\begin{aligned} (\gamma, \alpha(x + y)) &= (x + y, \gamma) = (x, \gamma)(y, \gamma) \\ &= (\gamma, \alpha(x))(\gamma, \alpha(y)) = (\gamma, \alpha(x) + \alpha(y)). \end{aligned}$$

Hence $\alpha(x + y) = \alpha(x) + \alpha(y)$, and α is a homomorphism. Since Γ separates points on G (Section 1.5.2), α is one-to-one, and so α is an isomorphism of G into $\hat{\Gamma}$.

The rest of the proof may be broken into three steps:

(a) α is a homeomorphism of G into $\hat{\Gamma}$.

(b) $\alpha(G)$ is closed in $\hat{\Gamma}$.

(c) $\alpha(G)$ is dense in $\hat{\Gamma}$.

Choose a compact set C in Γ , choose $r > 0$, and put

$$(1) \quad \begin{aligned} V &= \{x \in G: |1 - (x, \gamma)| < r \text{ for all } \gamma \in C\}, \\ W &= \{\hat{\gamma} \in \hat{\Gamma}: |1 - (\gamma, \hat{\gamma})| < r \text{ for all } \gamma \in C\}. \end{aligned}$$

By 1.5.2 and 1.2.6(c), these sets V form a neighborhood base at 0 in G , and the sets W form a neighborhood base at 0 in $\hat{\Gamma}$. The definition of α shows that

$$(2) \quad \alpha(V) = W \cap \alpha(G).$$

It follows that both α and its inverse are continuous at 0, and since α is an isomorphism, the same result holds, by translation, at any other point of G or of $\alpha(G)$.

7. The Pontryagin Duality Theorem (2)

It follows that both α and its inverse are continuous at 0, and since α is an isomorphism, the same result holds, by translation, at any other point of G or of $\alpha(G)$.

This proves step (a), and so $\alpha(G)$ is locally compact, in the relative topology which $\alpha(G)$ has as a subset of $\hat{\Gamma}$. Suppose $\hat{\gamma}_0$ is in the closure of $\alpha(G)$, and let U be a neighborhood of $\hat{\gamma}_0$ whose closure $\bar{U}$ is compact. Since $\alpha(G)$ is locally compact, $\alpha(G) \cap \bar{U}$ is compact, and hence closed in $\hat{\Gamma}$. But $\hat{\gamma}_0$ is in the closure of $\alpha(G) \cap \bar{U}$, and it follows that $\hat{\gamma}_0 \in \alpha(G)$. Thus $\alpha(G)$ is closed, and step (b) is proved.

If $\alpha(G)$ is not dense in $\hat{\Gamma}$, there is a function $F \in A(\hat{\Gamma})$ which is 0 at every point of $\alpha(G)$ but is not identically 0 (see Theorem 1.6.4). For some $\phi \in L^1(\Gamma)$, we have

$$(3) \quad F(\hat{\gamma}) = \int_{\Gamma} \phi(\gamma)(-\gamma, \hat{\gamma}) d\gamma \quad (\hat{\gamma} \in \hat{\Gamma}).$$

Since $F(\alpha(x)) = 0$ for all $x \in G$, it follows that

$$(4) \quad \int_{\Gamma} \phi(\gamma)(-x, \gamma) d\gamma = \int_{\Gamma} \phi(\gamma)(-\gamma, \alpha(x)) d\gamma = 0 \quad (x \in G),$$

and so $\phi = 0$, by the uniqueness theorem 1.3.6. Hence $F = 0$, by (3), and this contradiction proves step (c) and completes the proof.

1.7.3. Some consequences of the duality theorem. The symmetry between G and Γ which is now established shows that every theorem proved for the ordered pair (G, Γ) also holds for (Γ, G) , and this enables us to complete some of the results which were previously established in provisional form only.

(a) *Every compact abelian group is the dual of a discrete abelian group, and every discrete abelian group is the dual of a compact abelian group.* This follows from Theorem 1.2.5.

(b) *If $\mu \in M(G)$ and $\hat{\mu}(\gamma) = 0$ for all $\gamma \in \Gamma$, then $\mu = 0$.* This is the dual of Theorem 1.3.6.

(c) *$M(G)$ and $L^1(G)$ are semi-simple Banach algebras.* (See Appendix D5). Since the map $\mu \rightarrow \hat{\mu}(\gamma)$ is a complex homomorphism of $M(G)$, for each $\gamma \in \Gamma$, the semi-simplicity of $M(G)$ follows from the uniqueness theorem (b). The same uniqueness theorem evidently holds for $L^1(G)$, and so $L^1(G)$ is semi-simple.

(d) *If G is not discrete, then $L^1(G)$ has no unit. Hence $L^1(G) = M(G)$ if and only if G is discrete.*

For if G is not discrete, then Γ is not compact, by (a), and since $A(\Gamma) \subset C_0(\Gamma)$, $A(\Gamma)$ contains no non-zero constants, hence has no unit. Since $A(\Gamma)$ is isomorphic, as an algebra, to $L^1(G)$, the proof is complete.

(e) *If $\mu \in M(G)$ and $\hat{\mu} \in L^1(\Gamma)$, there exists $f \in L^1(G)$ such that $d\mu(x) = f(x)dx$, and*

7. The Pontryagin Duality Theorem (3)

$$(1) \quad f(x) = \int_{\Gamma} \hat{\mu}(\gamma)(x, \gamma) d\gamma \quad (x \in G).$$

By hypothesis, $\hat{\mu} \in L^1(\Gamma) \cap B(\Gamma)$; hence if f is defined by (1), the inversion theorem (applied to the pair (Γ, G) instead of (G, Γ)), shows that $f \in L^1(G)$ and

$$(2) \quad \hat{\mu}(\gamma) = \int_G f(x)(-x, \gamma) dx \quad (\gamma \in \Gamma).$$

Since $\hat{\mu}(\gamma) = \int_G (-x, \gamma) d\mu(x)$, the uniqueness theorem now implies that $d\mu = f dx$, and the proof is complete.

8. The Bohr Compactification (1)

1.8.1. Suppose Γ is the dual of the LCA group G , Γ_d is the group Γ with the discrete topology, and $\tilde{G}$ is the dual of Γ_d . Then $\tilde{G}$ is a compact abelian group which we call the *Bohr compactification* of G (Anzai and Kakutani [1]). Let β be the map of G into $\tilde{G}$ defined by

$$(1) \quad (x, \gamma) = (\gamma, \beta(x)) \quad (x \in G, \gamma \in \Gamma).$$

1.8.2. THEOREM. β is a continuous isomorphism of G onto a dense subgroup $\beta(G)$ of $\tilde{G}$.

This theorem allows us to regard G as a dense subgroup of $\tilde{G}$, so that $\tilde{G}$ is indeed a compactification of G . Note, however, that $\beta(G)$ is not a locally compact subset of $\tilde{G}$ and that β is not a homeomorphism, unless G is compact, in which case $G = \tilde{G}$ and $\Gamma = \Gamma_d$.

Proof: Since Γ separates points on G , β is one-to-one, and it is easy to verify, as in the beginning of the proof of the Pontryagin duality theorem, that β is an isomorphism.

Let W be a neighborhood of 0 in $\tilde{G}$. Since a subset of Γ_d is compact if and only if it is finite, Theorem 1.2.6 shows that there exist $\gamma_1, \dots, \gamma_n \in \Gamma$ and $r > 0$, such that W contains the set

$$\{\bar{x} \in \tilde{G}: |1 - (\gamma_i, \bar{x})| < r; i = 1, \dots, n\},$$

which is a neighborhood of 0 in $\tilde{G}$. Let

$$V = \{x \in G: |1 - (x, \gamma_i)| < r; i = 1, \dots, n\}.$$

Then V is a neighborhood of 0 in G , and $x \in V$ implies $\beta(x) \in W$. Thus β is continuous at 0, and hence at all points of G , by translation.

Finally, let H be the closure in $\tilde{G}$ of $\beta(G)$. If $H \neq \tilde{G}$, then $\tilde{G}/H$ is a non-trivial compact group, and hence there is a character ϕ on $\tilde{G}/H$ which is not identically 1. The map $\bar{x} \rightarrow \phi(\bar{x} + H)$ is then a continuous character on $\tilde{G}$, not identically 1, which is 1 if $\bar{x} \in H$. Consequently there exists $\gamma_0 \in \Gamma$, $\gamma_0 \neq 0$, such that $(x, \gamma_0) = (\gamma_0, \beta(x)) = 1$ for all $x \in G$. This last equation implies that $\gamma_0 = 0$, and this contradiction completes the proof.

1.8.3. We may interpret the theorem in the following way: G and Γ are given, G is the group of all continuous characters on Γ , $\tilde{G}$ is the group of *all* characters on Γ , and the fact that G (or $\beta(G)$) is dense in $\tilde{G}$ leads to an approximation theorem (Hewitt and Zuckerman [1]):

THEOREM. Given $\gamma_1, \dots, \gamma_n \in \Gamma$, given $\varepsilon > 0$, and given any character ϕ on Γ , there is a continuous character ψ on Γ such that

$$(1) \quad |\psi(\gamma_i) - \phi(\gamma_i)| < \varepsilon \quad (i = 1, \dots, n).$$

Proof: $\phi \in \tilde{G}$, and the set of all $\psi \in \tilde{G}$ satisfying (1) is open in $\tilde{G}$, hence intersects $\beta(G)$.

8. The Bohr Compactification (2)

1.8.4. A function f on a LCA group G is *almost periodic* if and only if it is a uniform limit of trigonometric polynomials on G . (This is not the usual definition, but is equivalent to it). The almost periodic functions on G are precisely those which have continuous extensions to $\tilde{G}$; in other words, they are the restrictions to G of the continuous functions on $\tilde{G}$, and so $\tilde{G}$ may also be obtained as the maximal ideal space of the Banach algebra whose members are the almost periodic functions on G (Loomis [1]). These relations between almost periodicity and $\tilde{G}$ are the reason for associating Bohr's name with $\tilde{G}$. We shall not use these relations and omit their proof.

9. A Characterization of $B(\Gamma)$ (1)

1.9.1. We recall that $B(\Gamma)$ is the set of all functions $\hat{\mu}$ on Γ which are Fourier-Stieltjes transforms of measures $\mu \in M(G)$. We norm $B(\Gamma)$ by defining $\|\hat{\mu}\| = \|\mu\|$.

We already know one characterization of $B(\Gamma)$: $\phi \in B(\Gamma)$ if and only if ϕ is a finite linear combination of continuous positive-definite functions. It seems difficult to apply this, however, whereas the following criterion will be very useful to us. It was proved by Bochner [2] on the real line; an integral analogue is due to Schoenberg [1]; for the general case, see Eberlein [1].

THEOREM. *Each of the following two statements about a function ϕ , defined on Γ , implies the other:*

(a) $\phi \in B(\Gamma)$ and $\|\phi\| \leq A$.

(b) ϕ is continuous, and

$$(1) \quad \left| \sum_{i=1}^n c_i \phi(\gamma_i) \right| \leq A \|f\|_{\infty}$$

for every trigonometric polynomial f on G , of the form

$$(2) \quad f(x) = \sum_{i=1}^n c_i(x, \gamma_i).$$

Proof: If (a) holds, then $\phi = \hat{\mu}$, $\|\mu\| \leq A$, and

$$(3) \quad \sum c_i \phi(\gamma_i) = \sum c_i \int_G (-x, \gamma_i) d\mu(x) = \int_G f(-x) d\mu(x).$$

Hence (a) implies (b).

To prove the converse, we pass to the Bohr compactification $\tilde{G}$ of G . In the notation of Section 1.8, the formula

$$(4) \quad f(\bar{x}) = \sum_{k=1}^n c_k(\gamma_k, \bar{x}) \quad (\bar{x} \in \tilde{G})$$

extends each trigonometric polynomial f on G to a trigonometric polynomial on $\tilde{G}$, and since G is dense in $\tilde{G}$, the norm $\|f\|_{\infty}$ is not altered by this extension. The linear functional T defined on the space of all trigonometric polynomials f of the form (4) by

$$(5) \quad Tf = \sum c_k \phi(\gamma_k)$$

thus satisfies the inequality

$$(6) \quad |Tf| \leq A \|f\|_{\infty};$$

thus T can be extended to a bounded linear functional on $C(\tilde{G})$, of norm not exceeding A , and the Riesz representation theorem implies that there is a measure $\mu \in M(\tilde{G})$ such that $\|\mu\| \leq A$ and

$$(7) \quad \sum c_k \phi(\gamma_k) = \int_{\tilde{G}} f(-\bar{x}) d\mu(\bar{x})$$

for all f of the form (4). Taking $f(\bar{x}) = (\gamma, \bar{x})$, for some $\gamma \in \Gamma$, we obtain

$$(8) \quad \phi(\gamma) = \int_{\tilde{G}} (-\gamma, \bar{x}) d\mu(\bar{x}) \quad (\gamma \in \Gamma).$$

9. A Characterization of $B(\Gamma)$ (2)

To complete the proof, we have to show that μ is concentrated on G (more precisely, on $\beta(G)$, in the notation of 1.8).

It follows from the Radon-Nikodym theorem (Appendix E9) that there is a Borel function g on $\tilde{G}$, of absolute value 1, such that $g d\mu = d|\mu|$, and since $C(\tilde{G})$ is dense in $L^1(|\mu|)$ (Appendix E8), there is a sequence of trigonometric polynomials f_n on $\tilde{G}$ such that

$$(9) \quad \lim_{n \rightarrow \infty} \int_{\tilde{G}} |f_n - g| d|\mu| = 0.$$

By (8), the transforms

$$(10) \quad \phi_n(\gamma) = \int_{\tilde{G}} (-\gamma, \tilde{x}) f_n(\tilde{x}) d\mu(\tilde{x}) \quad (\gamma \in \Gamma)$$

are finite linear combinations of translates of ϕ and hence are continuous on Γ (not merely on Γ_a !). By (9), $\{\phi_n\}$ converges uniformly to

$$(11) \quad \Phi(\gamma) = \int_{\tilde{G}} (-\gamma, \tilde{x}) d|\mu|(\tilde{x}) \quad (\gamma \in \Gamma),$$

and Φ is continuous on Γ . The representation (11) shows that Φ is positive-definite, and so, by Bochner's theorem, Φ is the Fourier-Stieltjes transform of a measure σ on the dual group G of Γ . The uniqueness theorem for Fourier-Stieltjes transforms now implies that $\sigma = |\mu|$, hence $|\mu|$ is concentrated on G , and so is μ .

1.9.2. THEOREM. *If $\phi_n \in B(\Gamma)$ and $\|\phi_n\| \leq A$ for $n = 1, 2, 3, \dots$, if $\phi \in C(\Gamma)$ and if*

$$(1) \quad \phi(\gamma) = \lim_{n \rightarrow \infty} \phi_n(\gamma) \quad (\gamma \in \Gamma),$$

then $\phi \in B(\Gamma)$ and $\|\phi\| \leq A$.

This is a corollary of Theorem 1.9.1.

Appendix

- A. Topology*
- B. Topological Groups*
- C. Banach Spaces*
- D. Banach Algebras*
- E. Measure Theory*

The appendices are short descriptions of those parts of topology, group theory, and functional analysis which are used in this book. They are intended to provide an easily available reference and to convince the reader that an acquaintance with only the more elementary parts of these subjects will give him the necessary prerequisites. They also contain a record of the exact forms in which certain concepts are used; this is significant in those cases in which the terminology is not standardized. Since most of this material is quite familiar, it seemed superfluous to document each theorem with a reference to a specific source. Every item may be found in at least one of the following well-known texts: Dunford and Schwartz [1], Halmos [1], Hille and Phillips [1], Kaplansky [2], Kelley [1], Loomis [1], Montgomery and Zippin [1], Pontryagin [1].

A. Topology (1)

A1. A family τ of subsets of a set S is called a *topology* on S if (a) S and the empty set belong to τ , (b) τ is closed under the formation of finite intersections and arbitrary unions. If a topology τ is defined on S , then S is called a *topological space* (it would be more accurate to reserve this name for the ordered pair (S, τ) , but we shall ignore this distinction, as well as similar ones that occur later) and the members of τ are called *open sets*; their complements are *closed*. The largest open set contained in a set $A \subset S$ is the interior of A . The smallest closed set containing A is the closure $\bar{A}$ of A . If B is the complement of A , then $\bar{A} \cap \bar{B}$ is the *boundary* of A . If $\bar{A} = S$, A is *dense* in S . If some countable set is dense in S , S is *separable*. If p is an interior point of A , then A is a *neighborhood* of p . The set whose only element is p is written $\{p\}$. If $\{p\}$ is open, then p is an *isolated point* of S . If $\{p\}$ is open for every $p \in S$, then S is a *discrete space*.

A2. A family Ω of open subsets of a topological space S is a *base* if every open subset of S is a union of sets belonging to Ω . A family Ω_p of neighborhoods of a point $p \in S$ is a *neighborhood base* at p if every neighborhood of p contains a member of Ω_p . If to every pair p_1, p_2 of distinct points of S there exist neighborhoods N_1, N_2 of p_1, p_2 which are *disjoint* (i.e., whose intersection is empty), then S is called a *Hausdorff space*.

A3. Any subset X of a topological space S is itself a topological space if the open sets of X are defined to be the intersections of the open sets of S with X . This topology is the *relative topology* induced in X by S .

A4. A subset A of S (the case $A = S$ is not excluded) is called *compact* if every family of open sets whose union contains A has a *finite* subfamily whose union contains A . If every point of S has a compact neighborhood, then S is *locally compact*.

Every closed subset of a compact space is compact. Every compact subset of a Hausdorff space is closed. In a locally compact Hausdorff space, every point has a neighborhood base which consists of compact sets.

If Ω is a family of compact sets with the *finite intersection property* (i.e., every finite subfamily of Ω has non-empty intersection), then the intersection of all members of Ω is non-empty.

A5. If S is a topological space, let ∞ denote a point not in S , put $S_\infty = S \cup \{\infty\}$, and call a subset A of S_∞ open either if A is an open subset of S or if the complement of A is a compact subset of S . Then S_∞ is a compact space, and is called the *one-point compactification* of S . If S is compact, $\{\infty\}$ is an isolated point of S_∞ . If S is a locally compact Hausdorff space, then S_∞ is a compact Hausdorff space.

A. Topology (2)

A6. A map f of a topological space X into a topological space Y is called *continuous* if $f^{-1}(E)$ is open in X for every open set E in Y ; here $f^{-1}(E)$ denotes the set of all $p \in X$ such that $f(p) \in E$. If K is compact, $K \subset X$, and f is continuous, then $f(K)$ is compact.

If $f(E)$ is an open subset of Y whenever E is an open set in X , then f is called an *open map*. If f is one-to-one, if $f(X) = Y$, and if both f and f^{-1} are continuous, then f is a *homeomorphism* of X onto Y .

A7. If f is a continuous open map of a locally compact Hausdorff space X onto a Hausdorff space Y , and if K is a compact subset of Y , then there is a compact subset C of X such that $K = f(C)$.

It follows from the hypotheses that there are finitely many points $p_1, \dots, p_n$ in X with compact neighborhoods $N_1, \dots, N_n$ such that $K \subset f(N_1) \cup \dots \cup f(N_n)$. Put $C = f^{-1}(K) \cap \bigcup_1^n N_i$. Since $f^{-1}(K)$ is closed and $\bigcup N_i$ is compact, C is compact.

A8. A set A in a topological space S is *connected* if it is not the union of two disjoint non-empty sets which are open in the relative topology induced in A by S . The *component* of a point $p \in S$ is the union of all connected subsets of S which contain p . Since the closure of a connected set is connected, components are closed sets. If no component of S contains more than one point, S is called *totally disconnected*.

A9. In a locally compact totally disconnected Hausdorff space, the compact open sets form a base.

A10. If τ and τ_1 are two topologies on a set S and if $\tau \subset \tau_1$, then τ is said to be *weaker* than τ_1 . This terminology does not exclude the case $\tau = \tau_1$.

If F is a family of maps of S into a topological space Y , the collection of all finite intersections of sets of the form $f^{-1}(V)$ ($f \in F$, V open in Y) forms a base for a topology τ_F on S . Each $f \in F$ is evidently continuous with respect to τ_F , and τ_F is the weakest topology on S with this property; τ_F is called the *weak topology induced in S by F* . Of particular importance is the case in which F is a collection of complex-valued functions (i.e., Y is the complex plane).

F is said to *separate points* (or to be *separating*) on S if to every pair of distinct points p_1, p_2 in S there corresponds an $f \in F$ such that $f(p_1) \neq f(p_2)$. If F separates points and if Y is a Hausdorff space, then the weak topology induced by F on S is also a Hausdorff topology.

A. Topology (3)

A11. If S is a topological space, $C(S)$ denotes the set of all bounded continuous complex-valued functions on S . The *support* of a complex function f on S is the closure of the set of all points p at which $f(p) \neq 0$. The set of all $f \in C(S)$ whose support is compact is denoted by $C_c(S)$.

If, for each $\varepsilon > 0$, the inequality $|f(p)| < \varepsilon$ holds for all p in the complement of some compact set, then f is said to *vanish at infinity*. The set of all $f \in C(S)$ which vanish at infinity is denoted by $C_0(S)$. Each $f \in C_0(S)$ may be extended to a continuous function on S_∞ by setting $f(\infty) = 0$. If S is compact, then $C(S) = C_0(S) = C_c(S)$.

A12. The spaces $C(S)$, $C_0(S)$, $C_c(S)$ are closed under pointwise addition, multiplication, and scalar multiplication: $(f + g)(p) = f(p) + g(p)$; $(fg)(p) = f(p)g(p)$; $(\alpha f)(p) = \alpha f(p)$. Since the usual commutative, associative, and distributive laws hold, these spaces are *algebras* (over the complex field). If we introduce a norm in $C(S)$ by setting

$$\|f\|_\infty = \sup_{p \in S} |f(p)| \quad (f \in C(S)),$$

the metric $\|f - g\|_\infty$ turns $C(S)$ and $C_0(S)$ into *complete* metric spaces, since they are also closed under the formation of limits of uniformly convergent sequences. In fact, $C(S)$ and $C_0(S)$ are simple examples of *Banach algebras* (Appendix D).

If S is a locally compact Hausdorff space, then $C_c(S)$ is dense in $C_0(S)$.

A13. Tietze's Extension theorem. This theorem is usually stated for real-valued functions, but the following equivalent formulation is better suited to our purpose; we recall that an *arc* is a homeomorphic image of a compact interval of the real line:

Suppose K is a compact subset of the locally compact Hausdorff space S , and f is a continuous map of K into an arc L . Then there exists a continuous map g of S into L such that $g(p) = f(p)$ for all $p \in K$.

A14. The Stone-Weierstrass theorem. *Let S be a locally compact Hausdorff space and let A be a subalgebra of $C_0(S)$ which separates points on S , which is self-adjoint (i.e., $f \in A$ implies $\bar{f} \in A$, where $\bar{f}$ is the complex conjugate of f) and which contains, for each $p_0 \in S$, a function f such that $f(p_0) \neq 0$. Then A is dense in $C_0(S)$.*

A. Topology (4)

A15. Suppose A is an index set, and S_α is a set, for each $\alpha \in A$. The *cartesian product* $S = \prod_{\alpha \in A} S_\alpha$ is the set of all p which are functions on A such that $p(\alpha) \in S_\alpha$, for all $\alpha \in A$; $p(\alpha)$ may be regarded as the α th coordinate of the point p . If A is finite, say $A = \{1, 2, \dots, n\}$, the notation $S = S_1 \times S_2 \times \dots \times S_n$ is also used for S , and the points of S may be regarded as n -tuples $p = (p_1, \dots, p_n)$ with $p_\alpha \in S_\alpha$.

Suppose now that each S_α is a topological space. For any finite choice of indices, say $\alpha_1, \dots, \alpha_n$, and for any choice of open sets $V_{\alpha_i} \subset S_{\alpha_i}$ ($1 \leq i \leq n$), let V be the set of all $p \in S$ such that $p(\alpha_i) \in V_{\alpha_i}$ ($1 \leq i \leq n$), and declare a subset E of S to be open if and only if it is a union of such sets V . Then S satisfies the axioms for a topological space, and is called the *topological product of the spaces S_α* .

Each $\alpha \in A$ can be regarded as a function on S whose value at a point $p \in S$ is $p(\alpha)$. If this is done, it becomes evident that *the topology of S is exactly the weak topology induced on S by A* .

If each S_α is a Hausdorff space, it is trivial that S is also a Hausdorff space. The analogous statement for compact spaces lies deeper:

THE TYCHONOFF THEOREM. *The topological product of any collection of compact spaces is compact.*

A16. A topological space is *metrizable* if its topology is induced by a metric. For a compact Hausdorff space S , the following three properties are equivalent:

- (a) S is metrizable;
- (b) S has a countable base;
- (c) $C(S)$ is separable.

A17. If S is a locally compact Hausdorff space, or if S is a complete metric space, the *Baire theorem* holds: *S is not the union of countably many closed sets, unless one of them contains a non-empty open set.*

B. Topological Groups (1)

Although many of the statements which follow apply to non-commutative groups as well as to commutative (abelian) ones, we shall confine our attention to the latter class.

B1. An *abelian group* is a set G in which a binary operation, $+$, is defined, with the following properties:

- (a) $x + y = y + x$ for all $x, y \in G$;
- (b) $x + (y + z) = (x + y) + z$ for all $x, y, z \in G$;
- (c) G contains an element 0 such that $x + 0 = x$ for all $x \in G$;
- (d) to each $x \in G$ corresponds an element $-x \in G$ such that $x - x = 0$. (We write $x - x$ in place of $x + (-x)$.)

If A and B are subsets of G , $A + B$ denotes the set of all elements of the form $a + b$, with $a \in A$, $b \in B$. Similarly, $-A$ is the set of all elements $-a$, where a ranges over A , and $A - B = A + (-B)$. If $x \in G$, it is customary to write $A + x$ instead of $A + \{x\}$. We call $A + x$ the *translate* of A by x .

A subset H of G which is itself a group, with respect to the same group operation, is a *subgroup* of G . For this it is necessary and sufficient that $H - H \subset H$. If $H \neq G$, then H is a *proper* subgroup of G . If $H = \{0\}$, then H is the *trivial* group.

B2. A *homomorphism* of a group G into a group G_1 is a map ϕ of G into G_1 such that

$$\phi(x + y) = \phi(x) + \phi(y) \quad (x, y \in G).$$

A homomorphism which is one-to-one is an *isomorphism*. If there is an isomorphism of G onto G_1 , then G and G_1 are *isomorphic groups*, and for many purposes one need not distinguish between them.

The *kernel* of a homomorphism ϕ is the set $\phi^{-1}(0)$; the kernel is always a subgroup.

If H is a subgroup of G , the sets $H + x$ ($x \in G$) are the *cosets* of H . Two cosets $H + x$ and $H + y$ are identical if and only if $x - y \in H$; otherwise, $H + x$ and $H + y$ are disjoint. The set of all cosets of H is denoted by G/H , and G/H becomes an abelian group (the *quotient group* of G modulo H) if we define

$$(H + x) + (H + y) = H + x + y \quad (x, y \in G).$$

The map $x \rightarrow H + x$ is a homomorphism of G onto G/H , with kernel H . It is called the *natural homomorphism* of G onto G/H .

Conversely, if ϕ is any homomorphism of G , the group $\phi(G)$ may be regarded as a quotient group of G : $\phi(G) = G/\phi^{-1}(0)$.

The *index* of a subgroup H of G is the number of elements of G/H ; it is either a positive integer, or infinite.

B. Topological Groups (2)

B3. If $x \in G$ and n is a positive integer, nx is the element $x + x + \dots + x$ (n summands). If $nx = 0$ for some n , the smallest positive integer with this property is the *order* of x ; if $nx \neq 0$ for all $n > 0$, then x has *infinite order*. If there is an integer q such that $qx = 0$ for all $x \in G$, then G is said to be of *bounded order*.

If $E \subset G$ and if no proper subgroup of G contains E , we say that G is *generated* by E , or that E is *a set of generators*. A group generated by *one* of its elements is *cyclic*.

B4. A *topological abelian group* is a Hausdorff space G which is also an abelian group, provided the map $(x, y) \rightarrow x - y$ is a continuous map of the product space $G \times G$ onto G . If, in addition, the topology of G is locally compact, then G is a *locally compact abelian (LCA) group*.

(The proof of Theorem B6 (with $H = \{0\}$) shows that the Hausdorff separation axiom can be replaced by the weaker requirement that every point be a closed set, without changing the class of groups so defined. But this is not important for our present purpose.)

It follows that the translation map t_x , defined by $t_x(y) = x + y$, is a homeomorphism of G onto G , for each $x \in G$, and so is the map $x \rightarrow -x$. If A is an open set of G and $B \subset G$, then $A + B$ is a union of translates of A and is therefore open. If A and B are compact, then $A + B$ is compact, being the image of the compact set $A \times B$ under the continuous map $(x, y) \rightarrow x + y$.

A set $E \subset G$ is *symmetric* if $E = -E$. Since $E \cap (-E)$ is symmetric, it follows that *in every LCA group G there is a neighborhood base at 0 which consists of compact symmetric sets*. Moreover, the continuity of addition shows that to every neighborhood W of 0 in G there corresponds a neighborhood V of 0 (which may be taken compact and symmetric) such that $V + V \subset W$.

B. Topological Groups (3)

B5. The closure of any subgroup of G is again a subgroup of G . Every closed subgroup of a LCA group is LCA. Every open subgroup is closed; this is so since every coset of an open subgroup H is open, and since H is the complement of the union of all but one of its cosets.

B6. THEOREM. *Suppose G is LCA, ϕ is the natural homomorphism of G onto G/H , where H is a closed subgroup of G , and a subset of G/H is declared open if and only if it is the image under ϕ of an open subset of G . Then G/H is a LCA group.*

Proof: By definition, ϕ is continuous and open, and hence G/H is locally compact. If $x, y \in G$ and $x - y \notin H$, there is a neighborhood W of 0 such that $x + W$ does not intersect $y + H$, since $y + H$ is closed. There exists a symmetric compact neighborhood V of 0 such that $V + V \subset W$, and for this V the sets $x + H + V$ and $y + H + V$ do not intersect. In other words, the points $x + H$ and $y + H$ of G/H have disjoint neighborhoods, and so G/H is a Hausdorff space. The continuity of the group operation in G/H is easily verified.

B7. If $\{G_\alpha\}$ is a collection of abelian groups, their *complete direct sum* is the group G defined as follows: G , as a set, is the cartesian product of the sets G_α , and addition is performed coordinatewise: if $x \in G$ and $y \in G$, $x + y$ is the element of G whose α th coordinate is $x(\alpha) + y(\alpha)$, in the notation of Appendix A15.

The *direct sum* of the groups G_α is the subgroup of their complete direct sum which consists of all x which have $x(\alpha) \neq 0$ for only finitely many α .

If we now introduce the product topologies, the following facts emerge, via the Tychonoff theorem:

The direct sum of any finite collection of LCA groups is a LCA group. The complete direct sum of any collection of compact abelian groups is a compact abelian group.

If $G = H_1 + H_2$, where H_1 and H_2 are subgroups of G , then G is (isomorphic to) the direct sum $H_1 \oplus H_2$ of these two subgroups if and only if $H_1 \cap H_2 = \{0\}$.

B. Topological Groups (4)

B8. THEOREM. *If G is an abelian group of bounded order, then G is a direct sum of cyclic groups.*

Proof: Each prime p has a largest power p^a ($a \geq 0$) which divides the order of some $x \in G$. Hence G contains elements x_p of order p^a . If $x^* = \sum x_{p_i}$, the sum being taken over the distinct primes p_i for which $a_i > 0$, then x^* has order $\prod p_i^{a_i}$. Thus G contains an element x^* whose order is a multiple of the order of every $x \in G$.

Suppose H is a proper subgroup of G and

(a) H is a direct sum of cyclic groups;

(b) if n is an integer and if $nx \in H$ for some $x \in G$, then $nx_0 \in H$ for some $x_0 \in H$.

The preceding paragraph, applied to G/H , shows that there exists $y^* \in G$ and an integer m such that $mx \in H$ for all $x \in G$ and such that $ry^* \notin H$ if $0 < r < m$. By (b), $my^* = my$ for some $y \in H$, and if $z = y^* - y$ then z has order m and the group K generated by H and z has property (a) and is larger than H .

Suppose $nx = y + tz$, where $x \in G$, $y \in H$, and n, t are integers. Let d be the greatest common divisor of m and n . Then $(mn/d)x \in H$, hence $(mt/d)z \in H$, hence m divides mt/d , and so d divides t . The congruence $ns \equiv t \pmod{m}$ is therefore solvable for s . Put $k = sz$. Then $nk - tz = (ns - t)z = 0$. Hence $n(x - k) = y = ny_0$, for some $y_0 \in H$, since (b) holds for H . We conclude: $nx = n(y_0 + sz)$. Thus (b) also holds for K .

By Zorn's lemma, there is an H which is maximal with respect to (a) and (b), and the above argument shows that then $H = G$.

B9. Since the topology of a topological abelian group G is translation invariant, it is easy to introduce the notion of *uniform continuity*: A map f from a subset E of G into a metric space with metric d is uniformly continuous on E if to every $\varepsilon > 0$ there exists a neighborhood V of 0 in G such that $d(f(x), f(y)) < \varepsilon$ whenever $x \in E$, $y \in E$, and $y - x \in V$.

THEOREM. *If f is a continuous map of the compact set E in G into a metric space, then f is uniformly continuous on E .*

Proof: Given $\varepsilon > 0$, there corresponds to each $x \in E$ a neighborhood W_x of 0 such that $d(f(x), f(y)) < \varepsilon/2$ if $y \in E \cap (x + W_x)$, and there are symmetric open neighborhoods V_x of 0 such that $V_x + V_x \subset W_x$. Since E is compact, there is a finite set of points $x_1, \dots, x_n$ in E such that the union of the sets $x_i + V_{x_i}$ covers E . If V is the intersection of these V_{x_i} , and $y - x \in V$, $x \in E$, $y \in E$, then $x \in x_i + V_{x_i}$ for some i , and $y \in x + V \subset x_i + V_{x_i} + V \subset x_i + W_{x_i}$. Hence

$$d(f(x), f(y)) < d(f(x), f(x_i)) + d(f(x_i), f(y)) < \varepsilon.$$

B10. One proves similarly that every $f \in C_0(G)$ is uniformly continuous on G if G is LCA.

C. Banach Spaces (1)

C1. A *normed linear space* X is a vector space over the complex field (i.e., an abelian group in which multiplication by complex numbers is also defined and satisfies the usual distributive laws) in which a non-negative real number $\|x\|$, the *norm* of x , is associated to each $x \in X$, with the following properties:

- (a) $\|x\| = 0$ if and only if $x = 0$;
- (b) $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in X$;
- (c) $\|\alpha x\| = |\alpha| \cdot \|x\|$ for all $x \in X$ and all complex numbers α .

If X is *complete* with respect to the metric defined by $d(x, y) = \|x - y\|$, i.e., if every Cauchy sequence in X converges, then X is called a *Banach space*.

The topology induced by the above metric is the *norm topology* of X . The set of all $x \in X$ with $\|x\| \leq 1$ is the *unit ball* of X .

C2. If M is a closed linear subspace of a normed linear space X , the quotient space X/M (see Appendix B2) becomes a normed linear space if we introduce the *quotient norm*

$$\|x + M\| = \inf_{y \in M} \|x + y\| \quad (x \in X).$$

If X is a Banach space, so is X/M .

C3. A map T of a normed linear space X into a normed linear space Y is a *linear transformation* if $T(x + y) = Tx + Ty$ and $T(\alpha x) = \alpha \cdot Tx$ for all $x, y \in X$ and all complex numbers α ; in other words, linear transformations are vector-space homomorphisms. The kernel of a linear transformation T is a linear subspace. T is said to be *bounded* if there is a real number C such that $\|Tx\| \leq C\|x\|$ for all $x \in X$; the smallest C with this property is the *norm* $\|T\|$ of T . Note that $\|T\| = \sup_{x \neq 0} \|Tx\|/\|x\|$.

A linear transformation T is bounded if and only if it is continuous. The set $L(X, Y)$ of all bounded linear transformations of X into Y is itself a normed linear space, and if Y is a Banach space, so is $L(X, Y)$.

If $T \in L(X_0, Y)$, where X_0 is a dense linear subspace of a normed linear space X and Y is a Banach space, then T has a unique extension to an element of $L(X, Y)$, with the same norm. This is a special case of the general metric space theorem which states that any uniformly continuous map into a complete space has a continuous extension to the completion of its domain.

C4. The complex field K , normed by the absolute value, is a Banach space. A bounded linear transformation of X into K is called a *bounded linear functional* on X , and $L(X, K)$ is the *dual space* of X , written X^* .

C. Banach Spaces (2)

C5. THE HAHN-BANACH THEOREM. *If M is a linear subspace (not necessarily closed) of the normed linear space X , and if S is a bounded linear functional on M , then there exists $T \in X^*$ such that $Tx = Sx$ for all $x \in M$, and such that $\|T\| = \|S\|$.*

COROLLARY. *If $x_0 \in X$ and x_0 is not in the closure of M , then there exists $T \in X^*$ such that $Tx = 0$ for all $x \in M$ but $Tx_0 \neq 0$.*

C6. The next theorem depends in an essential manner on the completeness of the spaces involved and is a consequence of the Baire theorem (Appendix A17):

THEOREM. *Suppose X, Y are Banach spaces, $T \in L(X, Y)$, T is one-to-one, and $TX = Y$. Then $T^{-1} \in L(Y, X)$.*

COROLLARY. *If a vector space X is a Banach space with respect to two norms, say $\|\cdot\|$ and $\|\cdot\|'$, and if there is a constant C such that $\|x\|' \leq C\|x\|$ for all $x \in X$, then there is a constant C' such that $\|x\| \leq C'\|x\|'$ for all $x \in X$.*

If these two inequalities hold, the two norms are called *equivalent*. A further consequence is

THE CLOSED GRAPH THEOREM. *If X and Y are Banach spaces, if T is a linear transformation of X into Y , and if the relations $\lim_{n \rightarrow \infty} \|x_n - x\| = 0$ and $\lim_{n \rightarrow \infty} \|Tx_n - y\| = 0$ imply that $y = Tx$, then T is bounded.*

We need only apply the preceding corollary to the map $x \rightarrow (x, Tx)$ of X onto the graph of T which is a Banach space with norm $\|x\| + \|Tx\|$.

C7. Each $x \in X$ may be regarded as a function on X^* whose value at a point $T \in X^*$ is Tx . Then X is a separating family of functions on X^* . The weak topology induced in X^* by X (Appendix A10) is called the weak* topology of X^* .

THEOREM. *For any normed linear space X , the unit ball S^* of X^* is a compact Hausdorff space in the weak* topology of X^* .*

Proof: Let D_x be the set of all complex numbers z with $|z| \leq \|x\|$, where $x \in X$. With X as index set, S^* is then a subset of the topological product D of the discs D_x (Appendix A15). By the Tychonoff theorem, D is a compact Hausdorff space, and since limits of linear functions are linear, S^* is a closed subset of D .

C. Banach Spaces (3)

C8. If X is separable, then a countable subset of X separates points on X^* . It follows that the weak* topology of X^* has a countable base. We conclude (see Appendix A16):

If X is separable, then the weak topology of the unit ball S^* of X^* is metrizable.*

C9. The following analogue of the corollary to the Hahn-Banach theorem is a direct consequence of the definition of the weak* topology:

If M is a weak-closed linear subspace of X^* and if $T_0 \notin M$, then there exists $x \in X$ such that $T_0 x \neq 0$ but $Tx = 0$ for all $T \in M$.*

C10. Suppose X and Y are Banach spaces, X^* and Y^* are their duals, and $T \in L(X, Y)$. For any $y^* \in Y^*$, the map $x \rightarrow y^*(Tx)$ is a bounded linear functional on X ; hence there is an element of X^* , which we write T^*y^* , such that $(T^*y^*)(x) = y^*(Tx)$ for all $x \in X$. The map T^* of Y^* into X^* so defined is called the *adjoint* of T . It is easy to see that $T^* \in L(Y^*, X^*)$.

C11. THEOREM. Suppose X, Y are Banach spaces, $T \in L(X, Y)$, T is one-to-one, and TX is dense in Y . Then each of the following three properties implies the other two:

- (a) $TX = Y$.
- (b) There exists $\delta > 0$ such that $\|T^*y^*\| \geq \delta\|y^*\|$ for all $y^* \in Y^*$.
- (c) $T^*Y^* = X^*$.

Proof: Let $S_r = \{x \in X: \|x\| \leq r\}$. If (a) holds, C6 shows $T(S_1)$ contains all $y \in Y$ with $\|y\| \leq \delta$. Hence

$$\|T^*y^*\| = \sup_{x \in S_1} |(T^*y^*)(x)| = \sup_{x \in S_1} |y^*(Tx)| \geq \delta\|y^*\|$$

for all $y^* \in Y^*$, and (b) holds. If (c) holds, one proves in the same way that $\|Tx\| \leq \alpha\|x\|$ for some $\alpha > 0$ and all $x \in X$; this implies that TX is norm-closed, and so (a) holds. It remains to show that (b) implies (c).

If (b) holds, then $\Gamma = T^*Y^*$ is norm-closed in X^* . Moreover, $T^{*-1}(E)$ is bounded for every bounded set E in Γ , and this implies, via C7, that the intersection of Γ with every closed ball in X^* is weak*-compact.

Suppose $\Gamma \neq X^*$. Then there exists $x_0^* \in X^*$ whose distance from Γ exceeds 1. Put $\Gamma_n = \{x^* \in \Gamma: \|x^* - x_0^*\| \leq n\}$, and if F is any finite subset of X , let F° be the set of all $x^* \in X^*$ such that $|(x^* - x_0^*)(x)| \leq 1$ for all $x \in F$.

C. Banach Spaces (4)

We claim that there exist finite sets $F_n \subset S_{1/n}$ such that

$$(1) \quad F_0^\circ \cap F_1^\circ \cap \dots \cap F_{n-1}^\circ \cap \Gamma_n \text{ is empty} \quad (n = 1, 2, 3, \dots).$$

Since Γ_1 is empty, F_0 may be chosen arbitrarily. Suppose (1) holds for some $n \geq 1$, $W_n = F_0^\circ \cap \dots \cap F_{n-1}^\circ$, and $F^\circ \cap W_n \cap \Gamma_{n+1}$ is not empty, no matter what finite set $F \subset S_{1/n}$ we take. Since $W_n \cap \Gamma_{n+1}$ is weak*-compact and since the collection of all sets $F^\circ \cap W_n \cap \Gamma_{n+1}$ has the finite intersection property, there exists $x^* \in W_n \cap \Gamma_{n+1}$ such that $|(x^* - x_0^*)(x)| \leq 1$ for all $x \in S_{1/n}$. But then $\|x^* - x_0^*\| \leq n$, or $x^* \in \Gamma_n$, which contradicts (1). Hence F_n exists, and the induction is complete.

Arrange the elements of $\bigcup F_n$ in a sequence $\{x_i\}$. Then $\|x_i\| \rightarrow 0$, and

$$(2) \quad \sup_i |(x^* - x_0^*)(x_i)| > 1 \quad (x^* \in \Gamma).$$

The map $\Psi: x^* \rightarrow \{x^*(x_i)\}$ is a bounded linear transformation of X^* into the space of all sequences which converge to 0, and (2) shows that the distance from $\Psi(x_0^*)$ to $\Psi(\Gamma)$ is positive. Hence there exists $\{\alpha_i\}$, with $\sum |\alpha_i| < \infty$, so that, setting $x = \sum \alpha_i x_i$, we have $x_0^*(x) = \sum \alpha_i x_0^*(x_i) = 1$ but $x^*(x) = \sum \alpha_i x^*(x_i) = 0$ if $x^* \in \Gamma$. The latter condition implies that $y^*(Tx) = (T^*y^*)(x) = 0$ for all $y^* \in Y^*$, so that $Tx = 0$ and hence $x = 0$, which is impossible if $x_0^*(x) = 1$. This contradiction proves that $\Gamma = X^*$, so that (c) holds.

C12. Suppose H is a vector space over the complex field, and suppose that to each ordered pair $x, y \in H$ there is associated a complex number (x, y) , called the *inner product* of x and y , with the following properties: (a) $(x_1 + x_2, y) = (x_1, y) + (x_2, y)$; (b) $(\alpha x, y) = \alpha(x, y)$; (c) $(y, x) = \overline{(x, y)}$; (d) $(x, x) \geq 0$; (e) $(x, x) = 0$ only if $x = 0$. Setting $\|x\| = (x, x)^{1/2}$, H then becomes a normed linear space. If H is complete in this norm, H is called a *Hilbert space*.

The *Schwarz inequality* $|(x, y)| \leq \|x\| \cdot \|y\|$ is a consequence of properties (a) to (d) of the inner product. It shows that the map $x \rightarrow (x, y)$ is, for each $y \in H$, an element of H^* . Conversely, to each $T \in H^*$ there corresponds a unique $y \in H$ such that $Tx = (x, y)$. Thus H is its own dual.

A set K in H is *convex* if $\alpha x + (1 - \alpha)y \in K$ whenever $x \in K$, $y \in K$, and $0 < \alpha < 1$. Each closed convex set K in a Hilbert space H has a unique element x_0 of minimal norm. If $x_0 + M \subset K$ for some linear subspace M of H , then $(x_0, y) = 0$ for all $y \in M$; in other words, x_0 is *orthogonal* to M . If 0 is the only element in H which is orthogonal to a linear subspace M , then M is dense in H , by the Hahn-Banach theorem and the above characterization of H^* .

D. Banach Algebras (1)

D1. A vector space A over the complex field is a *commutative algebra* if a multiplication is defined in A which satisfies the usual commutative, associative and distributive laws. If a norm is defined in a commutative algebra A which makes A into a Banach space, and if the inequality $\|xy\| \leq \|x\| \cdot \|y\|$ holds for all $x, y \in A$, then A is a *commutative Banach algebra*.

In this appendix, the symbol A will always denote a commutative Banach algebra.

There may or may not be a *unit* in A , i.e., an element e such that $xe = x$ for all $x \in A$. If A has a unit, the norm of A can be replaced by an equivalent one (see Appendix C6) such that $\|e\| = 1$. The element $x \in A$ is *invertible* if it has a multiplicative inverse, i.e., if there is an element $x^{-1} \in A$ such that $x^{-1}x = e$.

D2. A subalgebra I of A is an *ideal* in A if $xy \in I$ whenever $x \in A$ and $y \in I$. If $I \neq A$, I is a *proper* ideal. *Maximal ideals* are proper ideals which are not contained in any larger proper ideals.

THEOREM. *If A has a unit, then every proper ideal in A is contained in a maximal ideal, and every maximal ideal is closed.*

This is an easy consequence of Zorn's lemma and the following three facts: (a) proper ideals contain no invertible elements, (b) the set of all invertible elements is open, and (c) the closure of an ideal is an ideal.

D3. If I is an ideal in A , a multiplication may be defined in the quotient space A/I by setting

$$(x + I)(y + I) = xy + I \quad (x, y \in A);$$

this makes A/I into an algebra, the so-called quotient algebra of A modulo I .

If I is closed and if A/I is given the quotient norm (see Appendix C2), then A/I is a Banach algebra.

An ideal I in A is said to be *regular* if A/I has a unit (if A has a unit, *every* ideal is regular). Theorem D2 has the following replacement if A has no unit:

Every proper regular ideal in A is contained in a regular maximal ideal, and every regular maximal ideal is closed.

D. Banach Algebras (2)

D4. A *complex homomorphism* h of A is a linear functional on A which is also multiplicative: $h(xy) = h(x)h(y)$. Let Δ be the set of all complex homomorphisms of A which are not identically 0. The following statements contain the core of the theory of commutative Banach algebras, as developed by Gelfand [1]:

(a) If I is a regular maximal ideal in A , then A/I is (isometrically isomorphic to) the complex field, and so the canonical homomorphism of A onto A/I belongs to Δ .

(b) Conversely, if $h \in \Delta$, the kernel of h is a regular maximal ideal in A .

(c) If A has a unit, then $x \in A$ is invertible if and only if $h(x) \neq 0$ for all $h \in \Delta$. In any case, the equation $xy = x + y$ is solvable in A if and only if $h(x) \neq 1$ for all $h \in \Delta$.

(d) Each $h \in \Delta$ is a *bounded* linear functional on A , of norm 1. Thus Δ is a subset of the unit ball S^* in the dual space A^* of the Banach space A .

(e) Each $x \in A$ defines a function $\hat{x}$ on Δ , given by

$$\hat{x}(h) = h(x) \quad (h \in \Delta).$$

The weak topology induced in Δ by the collection of all these functions $\hat{x}$ is called the *Gelfand topology* of Δ . It coincides with the relative topology which Δ has as a subset of A^* if A^* is given the weak* topology. Since $\Delta \subset S^*$, since S^* is weak*-compact (Appendix C7) and since $\Delta \cup \{0\}$ is easily seen to be a closed subset of S^* , it follows that Δ is a locally compact Hausdorff space (usually called the *maximal ideal space* of A) and that each $\hat{x}$ is a member of $C_0(\Delta)$ (Appendix A11).

(f) The map $x \rightarrow \hat{x}$ is a homomorphism of A onto a subalgebra $\hat{A}$ of $C_0(\Delta)$, since

$$(\hat{x}\hat{y})(h) = h(xy) = h(x)h(y) = \hat{x}(h)\hat{y}(h) \quad (x, y \in A; h \in \Delta),$$

and similarly for addition and scalar multiplication. Since $\|h\| \leq 1$ the important inequality

$$\|\hat{x}\|_\infty \leq \|x\|$$

holds. We call $\hat{x}$ the *Gelfand transform* of x .

(g) If A has a unit e , then Δ is compact, since $\hat{e}(h) = h(e) = 1$ and $1 \in C_0(\Delta)$ only if Δ is compact.

D. Banach Algebras (3)

D5. If the Gelfand transformation is an isomorphism, i.e., if $x \neq 0$ implies $\hat{x} \neq 0$ (or, $h(x) \neq 0$ for some $h \in \Delta$), then A is said to be *semi-simple*.

THEOREM. If A and B are commutative Banach algebras, if B is semi-simple, and if Ψ is a homomorphism of A into B , then Ψ is continuous (i.e., $\Psi \in L(A, B)$). If $\Psi \neq 0$, then $\|\Psi\| \geq 1$.

Proof: Suppose $x_n \rightarrow x_0$ in A and $\Psi x_n \rightarrow y_0$ in B , for some sequence $\{x_n\}$ in A . For each $h \in \Delta$, the maximal ideal space of B , the map $x \rightarrow h(\Psi x)$ is a complex homomorphism α on A . By D4(d), h and α are continuous, so that

$$h(\Psi x_0) = \alpha(x_0) = \lim \alpha(x_n) = \lim h(\Psi x_n) = h(y_0).$$

Since B is semi-simple, we conclude that $y_0 = \Psi x_0$, and the continuity of Ψ follows from the closed graph theorem.

If $\Psi \neq 0$, the semi-simplicity of B implies that $h(\Psi x) \neq 0$ for some $x \in B$, $h \in \Delta$, and so $\alpha \neq 0$. By D4(d), $\|\Psi\| = \|h\| \cdot \|\Psi\| \geq \|\alpha\| = 1$.

D6. The *spectrum* of an element $x \in A$ is defined to be the range of the function $\hat{x}$ (with 0 adjoined if A has no unit, so that the spectrum is always a compact subset of the complex plane). The number $\|\hat{x}\|_\infty$ is the *spectral norm* or the *spectral radius* of x . The equation

$$(1) \quad \lim_{n \rightarrow \infty} \|x^n\|^{1/n} = \|\hat{x}\|_\infty \quad (x \in A)$$

is known as the *spectral radius formula*. For Fourier transforms, it was discovered by Beurling [1]. The general case is due to Gelfand [1].

Let α and β be the upper and lower limits of $\{\|x^n\|^{1/n}\}$. Since $|h(x)|^n = |h(x^n)| \leq \|x^n\|$ for all $h \in \Delta$, we have $\|\hat{x}\|_\infty \leq \beta$. If λ is not in the spectrum of x , D4(c) shows that there exists $y = y(\lambda) \in A$ such that $-\lambda y + xy = x$; also, $y(\lambda)$ is an analytic function of λ , outside the spectrum of x . If $|\lambda| = C > \|x\|$, then $y(\lambda) = -\sum_1^\infty (x/\lambda)^n$, so that

$$(2) \quad x^n = -\frac{1}{2\pi i} \int_{|\lambda|=C} \lambda^{n-1} y(\lambda) d\lambda \quad (n = 1, 2, 3, \dots).$$

If $\|\hat{x}\|_\infty < r < R$, the path of integration in (2) can be shrunk to the circle $|\lambda| = r$ without changing the integral. Hence $\|x^n/R^n\| \rightarrow 0$, $\alpha \leq R$, and so $\alpha \leq \|\hat{x}\|_\infty$. This proves (1).

D. Banach Algebras (4)

D7. A similar application of the Cauchy formula shows that analytic functions operate in Banach algebras:

Suppose A is a commutative-semi-simple Banach algebra, $x \in A$, and F is an analytic function defined on an open set which contains the spectrum of x ; if A has no unit, we require that $F(0) = 0$. Then there exists a unique $y \in A$ such that $\hat{y}(h) = F(\hat{x}(h))$ for all $h \in \Delta$.

E. Measure Theory (1)

E1. Our discussion will be confined to measures and integrals on locally compact Hausdorff spaces X . Let $\mathcal{B}$ be the smallest family of subsets of X which (a) contains all closed subsets of X , (b) is closed under the formation of countable unions, and (c) is closed under complementation. Then $\mathcal{B}$ is also closed under the formation of countable intersections. The members of $\mathcal{B}$ are called the *Borel sets* of X .

A *measure on X* is a set function μ , defined for all Borel sets of X , which is *countably additive* (i.e., $\mu(E) = \sum \mu(E_i)$ if E is the union of the countable family $\{E_i\}$ of pairwise disjoint Borel sets of X), and for which $\mu(E)$ is finite if the closure of E is compact.

With each measure μ on X there is associated a set function $|\mu|$, the *total variation* of μ , defined by

$$(1) \quad |\mu|(E) = \sup \sum |\mu(E_i)|,$$

the supremum being taken over all finite collections of pairwise disjoint Borel sets E_i whose union is E . Then $|\mu|$ is also a measure on X (Hewitt [1]). If

$$(2) \quad |\mu|(E) = \sup |\mu|(K) = \inf |\mu|(V),$$

for every Borel set E , where K ranges over all compact subsets of E and V ranges over all open supersets of E , then μ is called *regular*.

We put

$$(3) \quad \|\mu\| = |\mu|(X)$$

and define $M(X)$ to be the set of all complex-valued regular measures on X for which $\|\mu\|$ is finite.

It is clear that $M(X)$ is a normed linear space if addition and scalar multiplication are defined by

$$(4) \quad (\mu_1 + \mu_2)(E) = \mu_1(E) + \mu_2(E), \quad (\alpha\mu)(E) = \alpha \cdot \mu(E)$$

for every Borel set E and every complex number α .

We shall also consider *non-negative regular measures on X* ; for these, $+\infty$ is an admissible value.

E. Measure Theory (2)

E2. If μ is a measure on X and A is a Borel set, the *restriction* μ_A of μ to A is the measure defined by

$$(1) \quad \mu_A(E) = \mu(A \cap E).$$

If $\mu = \mu_A$, then μ is said to be *concentrated on* A . If two measures μ_1 and μ_2 are concentrated on disjoint sets, the pair (μ_1, μ_2) is said to be *mutually singular*; in that case

$$(2) \quad \|\mu_1 + \mu_2\| = \|\mu_1\| + \|\mu_2\|.$$

If $\mu \in M(X)$, then μ is concentrated on a σ -compact subset of X (i.e., on a set which is a countable union of compact sets) and among all closed subsets of X there is a smallest one, the *support* of μ , on which μ is concentrated.

Every $\mu \in M(X)$ has a unique decomposition of the form

$$(3) \quad \mu = \mu_1 - \mu_2 + i\mu_3 - i\mu_4$$

in which $\mu_j \geq 0$, $\mu_j \in M(X)$, and the pairs (μ_1, μ_2) and (μ_3, μ_4) are mutually singular. This is the *Jordan decomposition theorem*.

E3. A measure $\mu \in M(X)$ is called *discrete* if it is concentrated on a countable set; μ is *continuous* if $\mu(E) = 0$ for every countable set E . Every $\mu \in M(X)$ has a unique decomposition $\mu = \mu_d + \mu_c$, where μ_d is discrete and μ_c is continuous.

If $\mu \in M(X)$, if m is a non-negative measure on X , and if $\mu(E) = 0$ whenever $m(E) = 0$, then μ is said to be *absolutely continuous* with respect to m .

THE LEBESGUE DECOMPOSITION THEOREM. If $\mu \in M(X)$ and $m \geq 0$, then μ has a unique decomposition $\mu = \mu_a + \mu_s$, where μ_a is absolutely continuous with respect to m and μ_s is singular with respect to m .

E. Measure Theory (3)

E4. If $\mu \in M(X)$, the map $f \rightarrow \int_X f d\mu$ is a bounded linear functional on the Banach space $C_0(X)$ (see Appendix A11). The converse of this statement is

THE RIESZ REPRESENTATION THEOREM. *To each bounded linear functional T on $C_0(X)$ there corresponds a unique $\mu \in M(X)$ such that*

$$(1) \quad Tf = \int_X f d\mu \quad (f \in C_0(X)).$$

In this generality, the theorem was first proved by Kakutani [1]. Its history is discussed in Dunford and Schwartz [1], pp. 373, 380. Since (Hewitt [1])

$$(2) \quad \sup |Tf| = \|\mu\| \quad (f \in C_0(X), \|f\|_\infty \leq 1),$$

if T and μ are related by (1), we see that (1) establishes an isometric isomorphism between $M(X)$ and the dual of $C_0(X)$. In particular, $M(X)$ is a Banach space.

E5. Another useful version of the Riesz representation theorem is as follows (Halmos [1]):

To each linear functional T on $C_c(X)$ such that $Tf \geq 0$ if $f \geq 0$, there corresponds a unique regular non-negative measure m on X such that

$$Tf = \int_X f dm \quad (f \in C_c(X)).$$

E6. A complex function f defined on X is called a *Borel function* if $f^{-1}(V)$ is a Borel set for every open set V in the complex plane. If $\mu \in M(X)$, all bounded Borel functions on X are integrable with respect to μ , and the inequality

$$\left| \int_X f d\mu \right| \leq \|\mu\| \cdot \sup_{x \in X} |f(x)|$$

holds.

E. Measure Theory (4)

E7. If m is a non-negative measure on X and if $0 < p < \infty$, $L^p(m)$ denotes the set of all Borel functions f on X for which the norm

$$(1) \quad \|f\|_p = \left(\int_X |f|^p dm \right)^{1/p}$$

is finite. If we identify functions which differ only on a set E with $m(E) = 0$, $L^p(m)$ becomes a Banach space, normed by (1), if $1 \leq p < \infty$. $L^2(m)$ is a Hilbert space, with inner product $(f, g) = \int f \bar{g} dm$.

$L^\infty(m)$ is the space of all bounded Borel functions on X , normed by

$$(2) \quad \|f\|_\infty = \operatorname{ess\,sup}_{x \in X} |f(x)|;$$

the essential supremum of $|f|$ is, by definition, the smallest number λ such that $m(\{x : |f(x)| > \lambda\}) = 0$. Again we identify any two members f, g of $L^\infty(m)$ for which $\|f - g\|_\infty = 0$.

E8. If m is regular, then $C_c(X)$ is dense in $L^1(m)$. The set function μ defined by $\mu(E) = \int_E |f| dm$ belongs to $M(X)$. Hence, given $\varepsilon > 0$, there is a compact set E such that $\mu(E') < \varepsilon$, where E' is the complement of E . It follows that there is a bounded Borel function g , with compact support E , such that $\|f - g\|_1 < \varepsilon$.

A theorem of Lusin (Saks [1]) asserts that for every $\delta > 0$ there exists $h \in C_c(X)$ such that $h(x) = g(x)$ except possibly on a set S with $m(S) < \delta$. We may also take h so that $\|h\|_\infty \leq \|g\|_\infty$. Thus $\|g - h\|_1 < 2\delta\|g\|_\infty$, and hence $\|f - h\|_1 < \varepsilon$ if δ is small enough.

Essentially the same proof holds for $L^p(m)$, if $1 \leq p < \infty$.

E9. If $f \in L^1(m)$, the measure defined by $\mu(E) = \int_E f dm$ belongs to $M(X)$ and is absolutely continuous with respect to m . The converse of this proposition is

THE RADON-NIKODYM THEOREM. If $\mu \in M(X)$, if m is a non-negative measure on X , and if μ is absolutely continuous with respect to m , then there exists $f \in L^1(m)$ such that

$$\mu(E) = \int_E f dm$$

for all Borel sets E in X .

Also, $\|\mu\| = \int_X |f| dm = \|f\|_1$.

E. Measure Theory (5)

E10. Suppose $m \geq 0$, $1 < p < \infty$, and $1/p + 1/q = 1$. The bounded linear functionals T on $L^p(m)$ are in one-to-one correspondence with the members g of $L^q(m)$: each $T \in (L^p)^*$ is of the form

$$Tf = \int fg \, dm \quad (f \in L^p(m)),$$

and $\|T\| = \|g\|_q$. Thus $L^q = (L^p)^*$.

If X is the union of a disjoint family of σ -compact sets X_α such that each σ -compact subset of X is contained in the union of a countable subfamily of $\{X_\alpha\}$, then we also have $L^\infty = (L^1)^*$. *This condition is satisfied by every LCA group G .* For if V is a compact symmetric neighborhood of 0 in G , if $V_1 = V$ and $V_{n+1} = V_n + V$, then each V_n is compact, and $H = \bigcup_{n=1}^\infty V_n$ is a σ -compact open subgroup of G . The cosets of H have the properties required of the sets X_α in the preceding paragraph.

E11. Suppose μ and λ are regular measures on locally compact Hausdorff spaces X and Y . For any set $A \times B$ in $X \times Y$, where A and B are Borel sets in X and Y , respectively, define

$$(\mu \times \lambda)(A \times B) = \mu(A)\lambda(B).$$

The set function $\mu \times \lambda$ so defined on “rectangles” has a unique extension to a regular measure $\mu \times \lambda$ on the product space $X \times Y$.

FUBINI'S THEOREM. *If $\mu \geq 0$, $\lambda \geq 0$, f is a Borel function on $X \times Y$, and $f \geq 0$, then*

$$(1) \quad \begin{aligned} \int_{X \times Y} f d(\mu \times \lambda) &= \int_X \int_Y f(x, y) d\lambda(y) d\mu(x) \\ &= \int_Y \int_X f(x, y) d\mu(x) d\lambda(y). \end{aligned}$$

If $\mu \in M(X)$, $\lambda \in M(Y)$, f is a Borel function on $X \times Y$, and if

$$\int_X \int_Y |f(x, y)| d|\lambda|(y) d|\mu|(x) < \infty,$$

then (1) also holds.